GENERAL SCIENCE GRADE 10: ACIDS, BASES AND SALTS

CBE Phenomenon-Driven Lesson Sequence — Sub-Strand 2.4 (8 Lessons)

SUB-STRAND OVERVIEW	
Grade Level	10
Subject	General Science
Strand	Strand 2.0: Matter and Chemical Reactions
Sub-Strand	Sub-Strand 2.4: Acids, Bases and Salts
Total Duration	8 lessons × 40 minutes = 320 minutes total
Sub-Strand Content	– Acids and bases — meaning, identification using universal indicator and pH chart – pH levels of common substances in the environment (bleaching agents, vinegar, salts, soda, juices, water, wood ash, antacids, lemons) – Functions of acids and bases in biological and chemical processes (digestion, respiration, bleaching action) – Chemical properties of dilute acids and bases — reactions of acids with bases, acids with carbonates, and acids with metals – Salts — classification and behaviour when exposed to air (hygroscopy, deliquescence and efflorescence) – Applications of salts in real-life situations (agriculture, food industry, medicine, laundry, road use) – Effects of salts on the environment (eutrophication, soil and air pollution)
Learning Outcomes	By the end of the sub-strand, the learner should be able to: a) distinguish between acids and bases using the universal indicator and pH chart; b) explain the role of acids and bases in biological and chemical processes; c) identify products of chemical reactions of acids and bases; d) classify salts according to their behaviour when exposed to air; e) outline applications of salts in real-life situations; f) promote proper use of salts in day-to-day life.
Core Competencies	– Communication and Collaboration: as the learner sensitises the community on the acidity and basicity of substances used in day-to-day life such as bleaching agents, vinegar, salts, soda, juices, water, wood ash, antacids and lemons, and as the learner listens attentively to peers when discussing applications of salts. – Self-efficacy: as the learner creates awareness on the importance of using required amounts of salt on food to avoid cases of high blood pressure, and as the learner carries out experiments on chemical properties of dilute acids and bases and presents findings. – Digital Literacy: as the learner uses digital devices and print media to research on the applications of salts in day-to-day life.
Core Values	– Responsibility: as the learner carefully uses digital devices to research on the applications of salts in day-to-day life, and takes personal responsibility for safe laboratory practice during acid-base experiments. – Respect: as the learner respects each other's opinion while discussing with peers the meaning of acids, bases and salts, and listens attentively during plenary presentations.

Science & Engineering Practices	– Asking Questions and Defining Problems: as the learner formulates testable questions about why some foods and household substances in Kenya (lemon juice, wood ash, vinegar, soda) turn different colours on universal indicator paper. – Planning and Carrying Out Investigations: as the learner designs and conducts experiments to test the pH of environmental substances and to investigate the behaviour of salts (hygroscopy, deliquescence, efflorescence) when exposed to air. – Analysing and Interpreting Data: as the learner records and interprets colour changes on universal indicator and pH charts, and tabulates results from reactions of acids with bases, carbonates and metals. – Constructing Explanations and Designing Solutions: as the learner explains the role of acids and bases in biological processes (digestion, respiration) and proposes practical community solutions around correct salt use and environmental impact of eutrophication. – Obtaining, Evaluating and Communicating Information: as the learner uses digital and print media to research applications of salts, evaluates sources, and communicates findings to peers and community members.
Pertinent & Contemporary Issues (PCIs)	– Socio-economic and Environmental Education: as the learner discusses the effect of salts on the environment, including eutrophication, soil and air pollution from agricultural fertiliser overuse around farming areas such as the Rift Valley and Lake Victoria basin. – Safety and Security: as the learner carries out experiments and presents findings on chemical properties of dilute acids and bases, observing all laboratory safety precautions. – Learner Support Programme (Peer Education): as the learner discusses with peers the functions of acids and bases in biological and chemical processes — digestion, respiration and bleaching action.
Career Connections	– Food Scientist / Nutritionist: understanding pH and acid-base chemistry underpins food preservation, fermentation (yoghurt, uji, togwa) and quality control in Kenyan food industries. – Agricultural Extension Officer / Agronomist: knowledge of soil pH, fertiliser salts and their environmental effects is essential for advising farmers in counties such as Nakuru, Trans Nzoia and Uasin Gishu on crop production. – Pharmacist / Biomedical Scientist: acid-base chemistry is foundational to formulating antacids, intravenous fluids and medicines dispensed in Kenyan health facilities. – Environmental Scientist: understanding eutrophication and salt pollution is critical for protecting water bodies such as Lake Victoria and Lake Naivasha. – Chemical / Process Engineer: acid-base and salt reactions are core to industrial manufacturing — soda ash production at Magadi Soda Company is a direct Kenyan example.
Focus for Lessons	This unit investigates the chemistry of acids, bases and salts through the lens of everyday Kenyan substances — from lemon juice and wood ash to the soda ash mined at Lake Magadi. Learners develop skills in experimental investigation, data interpretation and community communication as they explore pH, acid-base reactions and the diverse behaviour of salts in both biological systems and the wider environment. The unit adopts a phenomenon-driven, inquiry-based approach in which every lesson phase is anchored to a compelling local anchoring phenomenon, building towards a cumulative explanatory model of how these chemical species interact and affect Kenyan life and livelihoods.
Driving Question / Key Inquiry Questions	DRIVING QUESTION: Why do some substances from our everyday Kenyan environment — lake water, lemon juice, wood ash, cooking salt, antacids — react so differently with metals, food and living things, and how can understanding their acid, base or salt chemistry help us use them safely and sustainably? KICD KEY INQUIRY QUESTIONS: 1. Why are salts important in day-to-day life? 2. How do salts behave when exposed to air?

Anchoring Phenomenon	ANCHORING PHENOMENON — "The Mystery of Lake Magadi's Soda Crusts and the Burning Ugali Pot" Students are shown two striking images side by side: (1) a photograph of Lake Magadi in the Rift Valley, where thick white crusts of trona (sodium sesquicarbonate) coat the shoreline and flamingos wade in caustic pink-red water — yet local Maasai communities have harvested these salts for centuries; and (2) a close-up photograph of the inside of an aluminium sufuria (cooking pot) used daily to boil uji (porridge) in a Nairobi household, showing a stubborn white mineral scale on the inside walls and a thin grey pitting on the outside where the pot was accidentally left with dilute lemon-water inside overnight. Both images are puzzling: the Lake Magadi water looks dangerously alkaline and yet it supports flamingo life and yields commercially valuable soda ash — how can something so caustic be so useful? The cooking pot was attacked by lemon juice (an acid) overnight, yet the same lemon juice is safe to drink. The mineral scale inside is left by hard water (dissolved mineral salts) — the same category of compounds as the trona at Magadi. Students are asked: What is happening chemically in both cases? Why do some substances burn metal while others form protective crusts? Why do some salts dissolve in water and others cake up or absorb moisture from the air? These questions anchor the entire unit and are returned to at the end of every lesson.
Storyline Thread	Lesson 1 — PREDICT & ANCHOR: Students are shown the Lake Magadi / sufuria phenomenon images. Without any instruction, they predict in writing what they think is happening chemically, sort a set of 12 common Kenyan substances (lemon, vinegar, wood ash, antacid, milk, uji, baking soda, bleach, salt, water, soda water, tomato juice) into three labelled cups — "acid / base / neither" — using only prior knowledge. Predictions are posted on the Driving Question Board (DQB). Teacher introduces the driving question and the unit storyline. Lesson 2 — OBSERVE (pH Investigation): Students test all 12 substances using universal indicator paper and the pH chart, recording colour changes and pH values in a results table. They compare actual pH results against their Lesson 1 predictions, note surprises, and update the DQB with new questions. Teacher consolidates the definition of acids and bases in terms of H^+ and OH^- ions, and students begin building their first explanatory model diagram. Lesson 3 — EXPLAIN (Biological & Chemical Roles of Acids and Bases): Using the antacid-and-stomach-acid supporting phenomenon, students model digestion as an acid environment and discuss why the small intestine requires a more basic pH. They examine bleaching action (sodium hypochlorite) and respiration (CO_2 dissolving to form carbonic acid in blood). Readings and peer discussion connect every role back to the anchoring phenomenon (caustic lake water supporting life; lemon juice attacking metal). Model diagrams are revised. Lesson 4 — OBSERVE & INVESTIGATE (Reactions of Acids with Bases, Carbonates and Metals): Students carry out three supervised experiments — (i) dilute HCl + NaOH with universal indicator (neutralisation); (ii) dilute H_2SO_4 + marble chips / eggshells (carbonates — linking to antacid phenomenon); (iii) dilute HCl + magnesium ribbon (metal reaction — linking to the pitted sufuria). Students write word equations and balanced chemical equations, identify products, and present findings in plenary. Safety protocols are explicitly rehearsed before each experiment. DQB is updated. Lesson 5 — EXPLAIN (Salts — Classification and Behaviour in Air): Using the canteen cook phenomenon (clumping salt in Mombasa rains), students are introduced to the three behaviours: hygroscopy, deliquescence and efflorescence. They carry out an investigation leaving small samples of NaCl, anhydrous $CaCl_2$, $Na_2CO_3 \cdot 10H_2O$ and $CuSO_4 \cdot 5H_2O$ in open petri dishes over 24 hours (set up at end of Lesson 4, observed at start of Lesson 5), recording mass changes and visual appearance. Students classify each salt and explain results using their model of ionic compounds and water interaction.

	Lesson 6 — EXPLAIN & APPLY (Applications of Salts in Real Life): Students research (digital devices or print media) and map applications of salts across five sectors — agriculture (CAN, DAP fertilisers in Rift Valley farms), food industry (table salt, preservatives), medicine (saline drips, antacids), laundry (washing soda), road use (salt gritting). They connect the Lake Magadi soda ash (Na_2CO_3) to industrial and domestic applications. A structured peer-teaching activity has groups teach one sector each. The community awareness project brief (correct salt intake and blood pressure) is introduced and students begin drafting their community sensitisation messages. Lesson 7 — DRIVING QUESTION BOARD SYNTHESIS & MODEL BUILDING: Students revisit every entry on the DQB posted since Lesson 1 and evaluate which questions can now be answered using accumulated evidence. Working in groups, they produce a revised cumulative model (poster or digital slide) that explains: (i) what acids, bases and salts are; (ii) how acids react with bases, carbonates and metals; (iii) how salts behave in air; (iv) how these phenomena connect to the Lake Magadi anchoring image and the pitted sufuria. Groups present models and offer peer critique. Teacher facilitates consensus on a class model. Lesson 8 — ENVIRONMENTAL IMPACT, COMMUNITY PROJECT & ASSESSMENT: Students investigate the eutrophication supporting phenomenon (Lake Victoria water hyacinth), linking excess fertiliser salts to environmental harm. They finalise and perform their community sensitisation presentations (on acidity/basicity of household substances and correct salt use). A written formative assessment uses the anchoring phenomenon images as stimulus: students explain the chemistry behind the Lake Magadi crusts, the pitted sufuria and the clumping canteen salt, demonstrating command of all sub-strand learning outcomes. Lesson closes with students writing one remaining question they still have on a sticky note for the DQB — modelling the ongoing nature of scientific inquiry.
Supporting Phenomena	– A farmer in Kericho notices that adding wood ash (an alkaline substance) to acidic red soil boosts his tea seedling survival rate — students investigate what wood ash does to soil pH and why acids and bases neutralise each other. – A nurse at a Kisumu clinic gives a patient an antacid tablet (calcium carbonate) for stomach acidity — students observe the fizzing reaction when the tablet dissolves and connect it to the reaction of carbonates with acids, explaining both the bubbles and the relief from pain. – A secondary school canteen cook in Mombasa notices that the salt she stores in an open container during the long rains becomes wet and clumps, while the salt stored in a sealed tin stays dry and free-flowing — students investigate hygroscopy, deliquescence and efflorescence to explain the difference. – Water hyacinth choking the shores of Lake Victoria near Kisumu has been linked partly to eutrophication from excess fertiliser salts (nitrates and phosphates) washing off farms into the lake — students research how over-application of agricultural salts causes environmental damage, connecting chemistry to real ecological crisis.

LESSON 1 (40 minutes): Predict and Anchor: What Is Happening at Lake Magadi and Inside the Sufuria?	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To activate prior knowledge about acids, bases and salts by engaging with a compelling Kenyan phenomenon and making evidence-free predictions that will be tested throughout the unit.
Knowledge	Students will be able to recall everyday Kenyan substances they associate with sourness, bitterness or neutrality, and will be able to state at least one chemical question raised by the Lake Magadi and sufuria phenomenon images.
Skills	Students will sort 12 common Kenyan substances into three categories using prior knowledge only, record written predictions with justifications, and collaboratively construct a Driving Question Board.
Attitudes	Students will demonstrate curiosity and intellectual courage by committing predictions to writing before receiving any instruction, and will show respect for peers whose predictions differ from their own.
Key Inquiry Question	Why are salts important in day-to-day life? (KICD Key Inquiry Question 1 — introduced here as an open puzzle; revisited and answered in Lessons 5, 6 and 8.)
Purpose in Storyline	This is the opening anchor lesson. Its sole purpose is to create cognitive dissonance — students notice that substances they use every day (lemon juice, wood ash, lake water, antacids) behave in surprisingly different ways toward metals, food and living things. All predictions posted today become the baseline the class will interrogate for the rest of the unit. No correct answers are given in this lesson; productive uncertainty is the goal.
Safety Notes	

B. LESSON OVERVIEW
Students arrive to find two large printed photographs displayed at the front of the room: (1) Lake Magadi with its white trona crusts and flamingos wading in pink-red alkaline water, and (2) a close-up of a household aluminium sufuria showing white mineral scale on the inside and pitting on the outside where lemon-water sat overnight. Without any teacher explanation, students write individual predictions about what is happening chemically. They then physically sort 12 laminated substance cards into three labelled cups or zones. Predictions are shared, posted on the class Driving Question Board, and the teacher formally introduces the driving question and the unit storyline. The lesson closes with students writing one burning question each on a sticky note for the DQB. No content is delivered; the entire 40 minutes is devoted to surfacing prior knowledge, generating productive questions and establishing the collaborative inquiry culture for the unit.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** [Watch Me] Source: TED-Ed Lessons	Students enter the room to find the two phenomenon photographs displayed prominently — no titles or captions are given, only the images. Each student receives a	Teacher stands at the door as students enter and says: Take your seat, look at the two photographs at the front, and do not touch anything on your desk	Elicit-before-instruct protocol. Students must commit their prior knowledge to paper and to a physical arrangement before any peer or teacher input. This	Teacher collects half-sheets at the end of the phase (or photographs student arrangements with a phone). These are NOT graded. They serve as a diagnostic

🔍 Search ARES for similar videos 📖 **HTML:** Properties of Acids (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	half-sheet of paper with three sentence starters: (1) I notice... (2) I wonder... (3) I think the chemistry happening here involves... Students spend 5 minutes writing individual responses in silence. They then receive a set of 12 laminated substance cards (lemon, vinegar, wood ash, antacid, milk, uji, baking soda, bleach, table salt, water, soda water, tomato juice). Three zones on the desk are labelled with folded card: ACID / BASE / NEITHER. Without talking, each student arranges the 12 cards into the three zones according to their own prior knowledge. They photograph or sketch their arrangement and write a one-sentence justification for at least three of their most uncertain placements. This individual silent phase takes 8 minutes total.	yet. When everyone is seated the teacher says: I am not going to explain these photographs. I want to know what YOU already think. Pick up your half-sheet and respond to the three sentence starters honestly. There are no wrong answers right now — I want your genuine thinking. WAIT TIME: Teacher is completely silent for 5 full minutes. Teacher then distributes substance cards and says: Now sort these 12 substances into the three zones on your desk using only what you already know. Do this alone and in silence. WAIT TIME: Teacher circulates without commenting for 4 minutes, noting interesting or surprising placements on a clipboard for later reference but making no corrections and giving no hints. Teacher then says: Write one sentence explaining why you placed your three most uncertain cards where you did.	surfaces naive conceptions (for example, many students place salt firmly in NEITHER without realising that dissolved salts can produce acidic or basic solutions) that will create productive cognitive dissonance throughout the unit. The phenomenon photographs function as an anchor — every subsequent lesson will return to these same images with new explanatory power.	baseline. Teacher notes: How many students correctly identify lemon and vinegar as acids? How many place wood ash or baking soda in BASE? How many place bleach correctly? How many place salt in NEITHER without hesitation? These data points guide the depth of consolidation needed in Lessons 2 and 3.
Observe Phase 📹 **VIDEO:** Watch Me Source: TED-Ed Lessons 🔍 Search ARES for similar videos 📖 **HTML:** Properties of Acids (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Students share their individual card arrangements with a partner seated beside them (Think-Pair protocol, 3 minutes). Each pair identifies their two biggest disagreements — substances they placed in different categories — and prepares to share one disagreement with the class. The teacher facilitates a 5-minute whole-class share-out. Each pair calls out one disagreement and the class shows by raised hands how many others had the same disagreement. The teacher tallies disagreements on the board under the heading: Things We Are Not Sure About. The class then views the two phenomenon photographs again as a projected image (if a projector is available) or as large	Teacher says: Turn to your partner. Compare your card arrangements. Find the two substances you placed differently. You have 3 minutes — go. WAIT TIME: 3 minutes of active pair discussion. Teacher then says: Each pair, tell me one substance you disagreed about and where each of you placed it. I am going to write these on the board. I will not tell you who is right yet. After tallying, teacher says: Look at this list. These are your genuine uncertainties. This is exactly what scientists feel at the start of an investigation. Now look at the photographs again. What do you see that you cannot explain? Write it on your sticky note.	The pair-share surfaces the most contested substances (typically bleach, milk, baking soda, and antacid) and makes disagreement socially normalised. The sticky-note observation task shifts students from classifying familiar objects to confronting genuinely unfamiliar phenomena (trona crusts, caustic pink water, pitted aluminium), which deepens the need-to-know that will drive inquiry in subsequent lessons.	Teacher reads sticky notes quickly as students write. Look for: Are students noticing surface features only (colour, appearance) or are some already invoking chemical language (acid, reaction, dissolve)? Students who use chemical language unprompted are identified as potential peer explainers in later lessons. Students who write only visual observations indicate they need more scaffolding in Lesson 2 to connect observation to inference.

	A3 prints passed around. Students are asked to write on a fresh sticky note: One thing I see in the Lake Magadi image that I cannot yet explain, and one thing I see in the sufuria image that I cannot yet explain.			
Explain Phase 📹 **VIDEO:** Watch Me Source: TED-Ed Lessons 🔍 Search ARES for similar videos 📖 **HTML:** Properties of Acids (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	This is NOT an explanation-by-teacher phase in Lesson 1. Instead, students attempt their own first explanations. Each student writes a 4-to-6-sentence paragraph titled: My Current Best Explanation — What I Think Is Happening Chemically at Lake Magadi and Inside the Sufuria. Students are explicitly told: Use whatever scientific words you already know. It is completely fine if your explanation is wrong. The goal is to write YOUR thinking as clearly as you can right now. Students share their paragraphs with their table group (groups of 4). One member of each group reads their paragraph aloud. The group listens and then identifies: one idea they all agree with, and one idea at least one person challenges. This takes 6 minutes total.	Teacher says: Now I want you to try to explain what is happening. Do not look anything up. Do not ask me. Write your best explanation using the science you already know from previous grades. You have 4 minutes to write. WAIT TIME: 4 minutes of silence. Teacher then says: Share your explanation with your table group. One person reads theirs aloud. After listening, agree on one idea you all accept and one idea someone wants to challenge. You have 2 minutes. After group share, teacher says: I heard some very interesting ideas. I am going to hold on to my explanations for now. What you have written today is your starting model. By the end of this unit you will come back to it and see how your thinking has grown.	The student-generated explanation creates an initial model that is deliberately imperfect. This is the first iteration of the explanatory model that students will revise in Lessons 3, 4 and 7. Framing it as a starting model rather than a wrong answer reduces anxiety and builds scientific identity. The group challenge step introduces peer critique as a normal part of science without teacher arbitration.	Teacher circulates and listens during group share. Key diagnostic questions: Are students using the word acid and base at all? Do any students mention ions, reactions or pH unprompted? Are students connecting the two images (seeing a link between the lake salt and the sufuria scale) or treating them as unrelated? These observations shape the depth of the pH investigation design in Lesson 2.
Driving Question Board (DQB) Creation 📹 **VIDEO:** Watch Me Source: TED-Ed Lessons 🔍 Search ARES for similar videos 📖 **HTML:** Properties of Acids (Flexbook) Source: CK-12 🔍 Search ARES	The teacher reveals a large empty board (a section of the classroom wall covered with paper, or a dedicated notice board) labelled: DRIVING QUESTION BOARD — Our Questions About Acids, Bases and Salts. The teacher reads the unit driving question aloud: Why do some substances from our everyday Kenyan environment — lake water, lemon juice, wood ash, cooking salt, antacids — react so differently with metals, food and living things, and how can understanding their acid, base or	Teacher says: This board is going to be the heart of our investigation for the next eight lessons. Every question you post today is a real scientific question. We will come back to this board at the end of every lesson. By Lesson 8, I want us to have answered most of these questions using evidence — not just by looking up answers, but by doing science. Teacher then says: Walk up and post your sticky notes. You have 2 minutes. After posting, teacher reads selected sticky notes aloud and says: I am	The DQB externalises student thinking and makes it a shared class resource. Grouping questions into clusters gives students an early map of the unit structure without delivering content. Returning to the DQB at the end of every lesson creates continuity and shows students that their questions have status. The board is also a formative assessment tool for the teacher — unanticipated questions reveal depth of prior knowledge or prior misconceptions.	Teacher photographs the completed DQB at the end of this lesson and again at the end of every subsequent lesson. Comparing photographs across lessons documents the evolution of student understanding. Questions that remain unanswered by Lesson 7 become discussion points in the synthesis lesson. Questions that are posted repeatedly by multiple students flag the most important conceptual targets for the unit.

for similar readings	salt chemistry help us use them safely and sustainably? Each student now takes their sticky note (from the Observe Phase) and adds it to the DQB. Students are then given one additional blank sticky note each and write: The ONE question I most want answered by the end of this unit. All sticky notes are posted. Teacher reads 5 to 6 aloud (choosing diverse examples) and groups them loosely into clusters: questions about reactions, questions about salts, questions about living things, questions about safety. The clusters are labelled lightly with a marker. Students see that their individual questions connect to the unit themes.	going to group these into clusters. Watch where your question lands. This is your research agenda. Teacher then says: Look at the cluster about reactions. Look at the cluster about salts. These are the exact topics in our sub-strand 2.4. Your questions are the same questions chemists ask.		
Model Building Phase 📹 **VIDEO:** Watch Me Source: TED-Ed Lessons 🔍 Search ARES for similar videos 📖 **HTML:** Properties of Acids (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	In the final 6 minutes, students open their exercise books and draw a simple two-panel sketch: Panel 1 — Lake Magadi (white crusts, pink water, flamingos) with arrows pointing outward labelled with their current best guess of what is in the water. Panel 2 — the sufuria (scale on the inside, pitting on the outside) with arrows labelled with their current best guess of what caused each feature. Under each panel they write: What I think it is + What I am not sure about. These sketches are kept in their books as Page 1 of their unit model — they will add to and revise this sketch in Lessons 3, 4 and 7. Students write the date prominently. The teacher explains the model-revision routine: every time we learn something new, we come back to this sketch and update it in a different colour so we can see our thinking change.	Teacher says: In your exercise books, draw two panels — the lake and the pot. Label what you think is happening using arrows. Write what you are sure about and what you are not sure about. This is Page 1 of your scientific model. You will revise it six more times before this unit ends. Teacher holds up a sample model drawn on the board (deliberately incomplete and containing one plausible-but-wrong label, for example labelling the lake water as simply dirty) and says: My model here is incomplete. Yours will be too. That is fine. Science starts with incomplete models. The goal is to keep improving it with evidence. WAIT TIME: 5 minutes for drawing. Teacher then says: In our next lesson, you will test all 12 substances with universal indicator paper and see how your predictions hold up. Keep your predictions safe — we will need	The two-panel sketch is the first iteration of the cumulative explanatory model that is the central sensemaking artefact of the entire unit. By physically reserving a page in their books and dating it, students experience the model as a living document rather than a finished product. The teacher modelling an incomplete model normalises productive uncertainty and positions revision as intellectual progress rather than correction of error.	Teacher asks students to hold up their open books and scans the room for: completeness of labelling (are arrows present?), use of chemical vocabulary (acid, base, salt, mineral, react), and honesty in the not-sure-about column (students who leave this blank may need encouragement to embrace uncertainty). No marks are given. Teacher says: I can see some of you have used words like acid and mineral — excellent scientific vocabulary. I can also see that most of us are genuinely not sure about a lot — that is exactly the right place to start.

		them.		

D. TEACHER REFLECTION
After this lesson, reflect on the following: (1) Which of the 12 substances caused the most disagreement in predictions — this tells you where the deepest misconceptions lie and where Lesson 2 must spend the most time. (2) Did students generate any questions about ions, pH or reactions spontaneously — if so, you may be able to move faster through the definitions phase in Lesson 2. (3) Was the DQB dominated by questions about one topic (for example all questions about reactions, none about salts) — if so, you may need a bridging prompt at the start of Lesson 5 to connect salts back to the phenomenon. (4) Did any student show discomfort with not being given the answers — note these students for additional reassurance in Lesson 2 about the inquiry process. (5) Were the two phenomenon photographs sufficiently engaging — if the Lake Magadi image did not generate curiosity, consider bringing in a short 2-minute video clip of the lake for Lesson 2 to deepen the anchor before the pH investigation.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Students observed two striking photographs: Lake Magadi with white trona crusts and pink-red alkaline water where flamingos live, and a household aluminium sufuria with white mineral scale inside and acid-pitting outside from lemon-water left overnight. They also observed that 12 familiar Kenyan substances are difficult to classify without evidence.
What did I learn?	Students have not yet learned any new content. They have surfaced what they already know (or believe they know) about acids, bases and salts from everyday Kenyan life and from previous grades. They have learned that their own prior knowledge produces disagreements that science can resolve.
How does this explain the phenomenon?	Students produced their first explanatory model: a two-panel sketch of the Lake Magadi and sufuria phenomena labelled with their current best guesses. These explanations are deliberately tentative and incomplete and will be revised across Lessons 2 through 7 as evidence accumulates.

LESSON 2 (80 minutes (2 × 40-minute lessons)): pH Investigation: Measuring the Acid-Base Character of Everyday Kenyan Substances	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To distinguish between acids and bases using universal indicator paper and the pH chart, and to begin explaining observed differences in terms of H^+ and OH^- ions.
Knowledge	Learners will be able to: (a) distinguish between acids and bases using the universal indicator and pH chart; (b) explain the role of acids and bases in biological and chemical processes; (c) define acids as substances that release H^+ ions in solution and bases as substances that release OH^- ions in solution.
Skills	Learners will: (i) correctly use universal indicator paper to test 12 substances and record colour changes against the pH chart; (ii) construct and interpret a results table comparing predicted and actual pH values; (iii) identify anomalies between predictions (Lesson 1) and observations and pose new questions; (iv) begin constructing a particle-level explanatory model diagram showing H^+ and OH^- ions in solution.
Attitudes	Learners will: appreciate the importance of evidence over intuition when predictions do not match observations; value precision and care when handling chemicals and reading pH colours; show curiosity about why Lake Magadi water and lemon juice — both described as 'dangerous' or 'caustic' — sit at opposite ends of the pH scale yet both affect metals and living things.
Key Inquiry Question	1. Why are salts important in day-to-day life? 2. How do salts behave when exposed to air?
Purpose in Storyline	In Lesson 1, students made predictions about where 12 substances — including Lake Magadi water, lemon juice, wood ash, cooking salt, and antacids — sit on the acid-base spectrum. In Lesson 2 (OBSERVE), students now gather actual evidence by testing all 12 substances with universal indicator paper, recording pH values, and comparing results against their Lesson 1 predictions. Surprises (e.g. cooking salt at pH 7, wood-ash solution strongly basic, Lake Magadi water at pH 9–11) generate new questions that are added to the DQB. The teacher consolidates the ionic definition of acids (H^+) and bases (OH^-), anchoring it to the phenomenon: the caustic pink water at Lake Magadi is highly basic (excess OH^- ions), while the lemon juice that pitted the aluminium sufuria overnight is acidic (excess H^+ ions). Students begin building their first particle-level model diagram — the foundation for every subsequent lesson in this sequence.
Safety Notes	Dilute acids (e.g. dilute hydrochloric acid or dilute lemon-juice concentrate) and dilute alkalis (e.g. dilute sodium hydroxide, wood-ash filtrate) are used. Learners must: wear safety goggles and avoid touching eyes; wash hands immediately after contact; NOT taste any substance; report spills to the teacher immediately. Broken glass or torn indicator paper should be disposed of in the designated waste container. The teacher should have clean water and paper towels at each bench. If wood-ash solution is used, remind learners it is a strong base and corrosive at high concentration — the school-prepared solution should be diluted to a safe working concentration before the lesson.

B. LESSON OVERVIEW

Lesson 2 is the primary OBSERVE lesson in the evidence-gathering sequence. Students conduct a structured pH investigation using universal indicator paper and a standard pH colour chart to test 12 substances drawn from the anchoring phenomenon and Kenyan everyday life: (1) Lake Magadi water (simulated with dilute sodium carbonate/bicarbonate solution), (2) lemon juice, (3) wood-ash filtrate, (4) cooking salt (NaCl) solution, (5) vinegar, (6) antacid tablet dissolved in water, (7) tap water (Nairobi), (8) sukari (sugar) solution, (9) dilute hydrochloric acid, (10) dilute sodium hydroxide, (11) milk, (12) baking soda solution. Students record colour changes, assign pH values, classify each substance as acid / base / neutral, and compare all results against their Lesson 1 predictions. The teacher then consolidates the ionic

definitions of acids and bases (H^+ / OH^-), links these definitions back to both images of the anchoring phenomenon, and students update the Driving Question Board. The lesson closes with students producing their first particle-level model diagram, sketching H^+ and OH^- ions distributed in lemon juice versus Lake Magadi water solutions.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Phase 1 — Predict (Revisit & Refine) [10 minutes] 📹 **VIDEO:** Experimental probability Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Scientific Investigation (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Students retrieve their Lesson 1 prediction tables (where they ranked the 12 substances from most acidic to most basic). Working in their bench groups of 4, they spend 3 minutes re-reading their predictions and annotating any they now feel uncertain about — without changing their original answers. Each group nominates one spokesperson to share the substance they are MOST uncertain about and why. Students write a one-sentence 'refined prediction' at the top of today's results table: 'I predict the pH of [substance] will be approximately ___ because ___.' This activates prior knowledge, creates cognitive investment in the upcoming evidence, and primes learners to notice surprises.	Teacher displays both anchoring-phenomenon images on the board (Lake Magadi shoreline; pitted aluminium sufuria) and says: 'Last lesson you made predictions about these 12 substances. Today is evidence day — we test every single one. Before we touch anything, spend 3 minutes looking at your Lesson 1 predictions. Put a star next to any substance you feel uncertain about.' [Allow 3 minutes silent individual review — no talking.] After 3 minutes, teacher cold-calls exactly 4 groups (one per bench): 'Group at the window bench — which substance are you most uncertain about, and what is your current best guess for its pH? Give me a number.' [10-second WAIT TIME before calling on any student.] After 4 groups respond, teacher writes the 4 'most uncertain' substances on the board in a column labelled WATCH THESE. Teacher says: 'These are your personal prediction checkpoints. By the end of this lesson you will know whether you were right or whether the evidence surprises you — and surprises in science are MORE valuable than being right.'	Strategy: Prediction Activation with Uncertainty Flagging. By asking students to identify the substance they are MOST uncertain about rather than the one they are confident about, the teacher exploits the 'desirable difficulty' principle — uncertainty creates deeper encoding when the answer is later revealed. The four flagged substances become a shared class focus during the Observe phase, giving students a concrete personal stake in the evidence.	Observable indicator: Every student has annotated their Lesson 1 table with at least one star before the teacher cold-calls. Teacher circulates during the 3-minute review — if a student's table is blank, teacher prompts: 'Which one feels like a guess to you?' The four cold-called responses are written on the board; teacher notes whether students' uncertainty clusters around salts (cooking salt, baking soda) or neutral substances — this reveals whether Lesson 1's conceptual framing of 'salts are neither acid nor base' was understood.
Phase 2 — Observe: pH Testing Investigation	Groups receive a practical tray containing: 12 labelled test tubes in a rack (each containing 5 ml of one of the 12 substances), a strip	Teacher demonstrates the technique for the whole class first (approx. 3 minutes): 'Watch how I dip the rod — I am not dipping the	Strategy: Structured Data Collection with Embedded Anomaly Detection. The 'Surprised? (Y/N)' column in the	Observable indicators: (1) All 12 rows of the Results Table are completed with a colour, a number, a classification, and a Y/N

[35 minutes] 📹 **VIDEO:** Experimental probability Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Scientific Investigation (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	of universal indicator paper torn into 12 small squares, a pH colour chart (laminated, one per group), a white ceramic tile for colour comparison, a pencil, and a pre-printed Results Table with columns: Substance | Predicted pH (from L1) | Indicator Colour Observed | pH Value Read | Classification (Acid / Base / Neutral) | Surprised? (Y/N). Students work systematically: dip one small square of indicator paper into a test tube using a glass stirring rod, lay the paper on the white tile, compare the colour to the pH chart under natural light, record the colour and read off the pH value, classify the substance, and mark Y/N in the 'Surprised?' column. Groups rotate through all 12 substances over approximately 25 minutes. For the final 5 minutes of this phase, groups tabulate their 'Surprised?' entries and identify their single biggest surprise to share in Phase 3. Students also sketch the colour of each strip in their exercise book beside the numerical pH, creating a colour-coded personal record. Connection to phenomenon: when students reach the 'Lake Magadi water' sample (dilute Na_2CO_3/$NaHCO_3$ solution), teacher narrates: 'This solution approximates the chemistry of the trona-saturated water at Magadi — record its pH carefully. And when you reach the lemon juice sample, think about what happened to the aluminium sufuria overnight.'	paper directly into the test tube, I am transferring one drop to the paper on the tile. Why do you think we do it this way? Think for 10 seconds.' [10-second WAIT TIME.] Cold-call 2 students. Expected answer: to avoid contaminating the substance / to keep the paper flat for accurate colour reading. Teacher validates and adds: 'Contamination would ruin every reading that follows — this is why scientists use clean rods each time.' During the 25-minute practical, teacher circulates with a checklist of 6 observable safety and technique indicators (goggles on; rod cleaned between samples; paper on white tile; colour matched under natural light; pH value recorded as a number not just a colour; 'Surprised?' column filled). Teacher pauses the class at the 12-minute mark with a 60-second whole-class interrupt: 'Hands up — who has tested the Lake Magadi sample? What colour did you get?' [Record 4–5 responses on the board.] 'Hands up — who is surprised by that result?' Then continues circulation. At the 20-minute mark: 'You should have at least 8 substances completed. If you have fewer than 8, focus on the starred substances from Phase 1 first.' Teacher specifically visits any group that has recorded pH 7 for lemon juice and probes: 'Look at the colour again — hold it against the chart under the window light. What does that colour say to you?'	results table is not cosmetic — it is a deliberate sensemaking scaffold. By requiring students to flag surprises in real time (rather than in retrospect), the teacher ensures that cognitive dissonance is registered at the moment of observation, making it available for explanation in Phase 3. The mid-practical whole-class interrupt on the Lake Magadi sample creates a shared reference point: the entire class now has a common datum (pH ≈ 9–11) anchored to the phenomenon image.	entry by the end of 25 minutes. (2) Teacher's circulating checklist: at least 5 of 6 technique indicators met per group. (3) The mid-practical interrupt reveals whether the class recognises Lake Magadi water as basic (pH > 7). RED FLAG: if more than 3 groups record Lake Magadi water as pH 7 or below, the teacher must stop the class, re-demonstrate colour matching, and have all groups re-test that sample before proceeding. (4) At the end of Phase 2, teacher asks groups to report their count of 'Surprised?' entries — expected range is 2–5 surprises per group; fewer than 2 may indicate students are guessing their recorded values to match predictions.
Phase 3 — Explain: From Colour to Chemistry [20 minutes]	Part A — Sharing Surprises (7 minutes): Each group reports their single biggest surprise. Students listen and add a tick beside any surprise they also experienced.	For the Surprise Tally, teacher calls on one group per bench (4 groups) and writes their surprise on the board with a tally counter. Teacher says: 'Who else was	Strategy: Claim-Evidence-Reasoning (CER) anchored to Particle Diagrams. The teacher's board diagram makes the abstract ionic explanation concrete and	Observable indicators: (1) Every student's Results Table has an ion label (H^+ or OH^-) beside each acid/base classification. (2) The three cold-called students can

📹 **VIDEO:** [Experimental probability] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Scientific Investigation (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	The class builds a shared 'Surprise Tally' on the board. Likely top surprises: cooking salt (NaCl) at exactly pH 7 — many students predicted it would be acidic because it 'tastes salty'; wood-ash solution at pH 9–11 — many predicted it would be neutral; baking soda at pH 8–9 — some predicted it as neutral. Part B — Teacher-led Consolidation (10 minutes): Teacher introduces the ionic definitions of acids and bases using a worked particle diagram on the board: 'An acid releases hydrogen ions — H^+ — when dissolved in water. A base releases hydroxide ions — OH^-. The more H^+ ions, the lower the pH. The more OH^- ions, the higher the pH. pH 7 is neutral — equal H^+ and OH^-.' Students annotate their Results Table with the ion type beside each classification. Teacher explicitly links back to phenomenon: 'The Lake Magadi water has a very high pH because the trona — sodium sesquicarbonate — releases hydroxide ions. That is why the Maasai call it 'burning water' and why flamingo legs are eroded by it. The lemon juice that attacked the aluminium sufuria has a very low pH because citric acid releases H^+ ions — those hydrogen ions reacted with the aluminium metal overnight.' Part C — Application Question (3 minutes): Students answer individually in their books: 'A Nairobi market vendor sells wood ash as a cleaning agent for pots. Using pH evidence from today's investigation, explain ONE reason why wood ash might be effective at cleaning a greasy pot, and ONE reason why it might be dangerous if used without rinsing.'	surprised by cooking salt being exactly pH 7? Raise your hand.' [Count hands, record number.] This technique (named Surprise Tally) makes individual surprise publicly visible and validates cognitive dissonance as scientifically meaningful rather than as a mistake. For the consolidation, teacher draws a simple beaker diagram on the board: on the left side labels it 'LEMON JUICE — pH 2' and draws small '+' symbols (H^+ ions); on the right side labels it 'LAKE MAGADI WATER — pH 10' and draws small '–' symbols (OH^- ions). Teacher says: 'Copy this diagram into your exercise book. I am going to give you 30 seconds to label the ions, then I will cold-call three people.' [30 seconds.] Cold-calls 3 students by name (not volunteers): 'Amina — what ion makes lemon juice acidic?' / 'Kamau — what does a high pH number tell us about which ions dominate?' / 'Wanjiku — which end of the pH scale would the water inside the aluminium sufuria sit on after the lemon juice was left overnight?' [10-second WAIT TIME before each cold-call response.] For Part C, teacher says: 'Work alone for 3 minutes — I want a written answer, not a spoken one, before we share.' [3 minutes silent individual writing.] Teacher circulates, reads 3–4 student answers, selects one strong answer and one partial answer to share anonymously in whole-class discussion.	visual. By asking students to complete their own copy and be cold-called on it, the strategy moves from 'teacher explains' to 'student retrieves and applies.' The Part C application question requires students to construct a CER argument (Claim: wood ash is effective; Evidence: pH 9–11 = OH^- ions; Reasoning: OH^- ions can break down fats/grease in saponification), connecting the day's evidence to the anchoring phenomenon context of everyday Kenyan kitchen chemistry.	correctly name the relevant ion and connect it to pH value. RED FLAG: if any cold-called student says 'H^+ makes it basic,' teacher immediately re-draws the diagram and repeats the explanation before moving on. (3) Part C written answers — teacher collects 6 books (one from each side of each bench) to read during the DQB phase; target: at least 4 of 6 include both a valid reason for effectiveness AND a valid safety concern referenced to pH evidence.

	Students share with a partner before whole-class cold-call.			
Phase 4 — Driving Question Board (DQB) Update [8 minutes] 📹 **VIDEO:** Experimental probability Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Scientific Investigation (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Each student receives two sticky notes (one yellow, one blue). On the YELLOW note, they write one NEW question generated by today's evidence — something they did not know they wanted to ask before testing. On the BLUE note, they write one question from the Lesson 1 DQB that today's investigation has now ANSWERED (at least partially). Students post both notes on the class DQB wall display. Teacher reads 4–5 yellow notes aloud and sorts them into clusters: 'Questions about WHY pH matters for living things,' 'Questions about what happens when acid meets metal,' 'Questions about salts.' The teacher circles any DQB question that connects directly back to the anchoring phenomenon and labels it with a flame symbol (🔥) to signal: 'This is a question the whole unit is trying to answer.' Students update their personal DQB section in their exercise book — a running list of questions that will be revisited at the end of every lesson.	Teacher says: 'You have 2 minutes — two sticky notes, one new question, one answered question. Go.' [Strict 2-minute timer.] While students write, teacher prepares 4 blank cluster labels on the DQB: 'Living things & pH,' 'Acids + metals,' 'Salts,' 'Other.' After posting, teacher reads 5 yellow notes aloud, asks the class to vote by show of hands on which cluster each belongs to. Teacher explicitly narrates the connection to the phenomenon for at least 2 notes: e.g. 'This question — why does Lake Magadi water not kill ALL the flamingos if it is so basic — is a 🔥 question. It connects to our anchoring phenomenon and we will come back to it in Lesson 5 when we study salts and their effects on living things.' Teacher finishes by pointing to the original phenomenon images: 'Look at the sufuria image. After today, can anyone now explain in one sentence what the lemon juice was doing to the aluminium overnight?' Cold-call 2 students. [10-second WAIT TIME each.] Expected answer: 'The H^+ ions from the citric acid in lemon juice reacted with the aluminium metal.'	Strategy: Two-Colour DQB Update (New Questions / Answered Questions). The two-colour system makes epistemic progress visible — students must actively distinguish between 'what I now understand' and 'what I still do not understand.' This metacognitive routine, run at the end of every lesson in the sequence, trains students to track their own knowledge growth across the unit, which is a core CBE self-efficacy competency. The teacher's flame symbol (🔥) creates a visual hierarchy on the DQB that prioritises phenomenon-connected questions over peripheral ones.	Observable indicators: (1) Every student posts both a yellow and a blue note — teacher scans the board; any student who has not posted is prompted immediately. (2) Yellow notes are genuine questions (not statements dressed as questions); teacher reads them and flags any that are actually statements for the student to reframe. (3) The two cold-called students can connect lemon juice to H^+ ions and aluminium corrosion without teacher prompting. (4) Teacher notes the distribution of yellow-note questions across clusters — if fewer than 3 yellow notes mention salts, teacher adds a teacher-generated prompt note: 'We tested cooking salt (NaCl) today — it was pH 7. But is it really chemically inert? What happens to it when exposed to air?' This seeds the inquiry for Lesson 3 (hygroscopy, deliquescence, efflorescence).
Phase 5 — Model Building [7 minutes] 📹 **VIDEO:** Experimental probability Source: Khan Academy (English - US curriculum) 🔍 Search ARES	Students open a fresh page in their exercise books labelled 'MY EXPLANATORY MODEL — Version 1.' They draw two side-by-side beaker diagrams: Beaker A — lemon juice (pH 2); Beaker B — Lake Magadi water (pH 10). Inside each beaker, they draw and label: water molecules (H_2O), the dominant ion type (H^+ for Beaker	Teacher projects a partially completed model diagram (Beaker A only, with H_2O molecules drawn but no ions labelled) and says: 'I have started Beaker A. You have 5 minutes to complete both beakers. Use your Results Table and the board diagram from Phase 3.' [5-minute silent individual work — no group discussion.] Teacher	Strategy: Progressive Model Building (Version-Numbered Models). Requiring students to number and date each model version is a direct implementation of NGSS Science and Engineering Practice 2 (Developing and Using Models). Version-numbering makes model revision — rather than model replacement — the	Observable indicators: (1) Both beakers are drawn, labelled with the correct dominant ion (H^+ for lemon juice, OH^- for Lake Magadi water), and captioned. (2) The source molecule is named in at least one beaker. (3) The two cold-called students can describe their model verbally without reading from the drawing — demonstrating

for similar videos 📖 **HTML:** Scientific Investigation (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	A; OH^- for Beaker B), and the source molecule that released those ions (citric acid for Beaker A; sodium sesquicarbonate / trona for Beaker B). Students add a caption below each beaker: 'This substance is [acidic/basic] because it has more [H^+/OH^-] ions than the other.' Students date and sign their model. Teacher tells the class: 'This is Version 1. You will revise this model in every lesson — by Lesson 8, it will be much more detailed and powerful. Keep every version so you can see your own thinking grow.' The model is explicitly connected to the phenomenon: 'You are now beginning to explain, at the particle level, WHY the water at Lake Magadi corrodes flamingo legs AND why the lemon juice pitted the aluminium sufuria. Same logic, opposite chemistry.'	circulates and provides two specific verbal prompts only — no drawing for students: (1) If a student has drawn OH^- in the lemon juice beaker: 'Check your Results Table — what was the pH of lemon juice? What does a pH below 7 tell you about which ion dominates?' [10-second WAIT TIME.] (2) If a student has not labelled the source molecule: 'Where do the H^+ ions come from? What substance was dissolved in the lemon juice?' At the 5-minute mark, teacher cold-calls 2 students to describe their model aloud — not to show it. 'Describe Beaker A to me in two sentences without showing me the drawing.' This forces verbal articulation of the model, which is a higher-order sensemaking demand than drawing alone. Teacher closes: 'Date your model. Label it Version 1. Next lesson, you will add to it — do NOT erase anything. Scientists never erase early models; they revise them.'	norm, which mirrors authentic scientific practice. The constraint of silent individual work (no group discussion) in this phase ensures that the model represents each student's personal understanding, making it a valid formative assessment artefact rather than a copied group product.	conceptual (not just representational) understanding. RED FLAG: if a cold-called student describes both beakers as having the same ions, teacher asks the whole class: 'What is the difference between a pH of 2 and a pH of 10? Turn and tell your partner — 20 seconds.' Then re-cold-calls. (4) Teacher collects 4 model books (two from each bench side) to review overnight for Version 1 misconceptions to address at the start of Lesson 3.

D. TEACHER REFLECTION

After the lesson, reflect on the following questions in your professional journal:

1. EVIDENCE QUALITY: Did all groups complete all 12 rows of the Results Table with a colour, pH number, classification, and Y/N surprise entry? If not, which substances were most often skipped or incorrectly classified — and why? (Common issue: baking soda and cooking salt are often both recorded as pH 7 because the indicator colour difference between pH 7 and pH 8–9 is subtle in fluorescent classroom lighting. Consider having groups re-test these two side by side under a window for Lesson 3.)

2. SURPRISE DISTRIBUTION: Which substance generated the most 'Surprised?' responses across the class? Was it cooking salt (pH 7), wood ash (strongly basic), or Lake Magadi water? The surprise pattern reveals what prior conceptions students brought from Junior School — document this for your Lesson 3 planning.

3. IONIC DEFINITIONS: After the Phase 3 consolidation, how many of the cold-called students could correctly name H^+ as the acid ion and OH^- as the base ion WITHOUT prompting? If fewer than 2 out of 3 could do so, the ionic definition needs reinforcement at the START of Lesson 3 before moving to acid-base reactions.

4. MODEL VERSION 1: From the 4 books collected at the end of Phase 5, note: (a) Are the ions drawn in the correct beaker? (b) Is the source molecule named? (c) Are water molecules included (showing the dissolution context)? Models that show only ions floating in empty space — without water molecules — indicate students have not yet connected pH to aqueous solution. Address this in Lesson 3.

5. DQB QUALITY: Count the yellow notes that directly reference the anchoring phenomenon (Lake Magadi / the sufuria). If fewer than 30% of yellow notes reference the phenomenon, the lesson may have drifted into generic acid-base testing disconnected from the driving question. Strengthen the phenomenon narration during the

practical phase in future lessons. 6. SAFETY: Were all students wearing goggles throughout Phase 2? Was there any skin contact with wood-ash solution or dilute HCl? Document any incident, however minor, and report per school protocol. Consider whether the wood-ash solution concentration needs adjustment before Lesson 3.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Students tested 12 substances — including Lake Magadi water (simulated), lemon juice, wood-ash filtrate, cooking salt solution, vinegar, antacid solution, tap water, sugar solution, dilute HCl, dilute NaOH, milk, and baking soda solution — using universal indicator paper and a pH colour chart. They recorded indicator colour, pH value, and acid/base/neutral classification for each substance, and flagged surprises by comparing actual pH results against their Lesson 1 predictions.
What did I learn?	Substances are classified as acids (pH < 7, dominant H^+ ions), bases (pH > 7, dominant OH^- ions), or neutral (pH = 7, equal H^+ and OH^-). Lake Magadi water is strongly basic (pH 9–11) because dissolved trona (sodium sesquicarbonate) releases OH^- ions — explaining why it corrodes flamingo legs and aluminium pots. Lemon juice is acidic (pH 2–3) because citric acid releases H^+ ions — explaining why it pitted the aluminium sufuria overnight. Cooking salt (NaCl) solution is neutral (pH 7). Wood-ash solution is basic (pH 9–11). Evidence does not always match intuitive predictions — surprises are more scientifically valuable than confirmations.
How does this explain the phenomenon?	Acids release hydrogen ions (H^+) when dissolved in water; the more H^+ ions, the lower the pH and the more acidic the substance. Bases release hydroxide ions (OH^-); the more OH^- ions, the higher the pH and the more basic the substance. The pH scale runs from 0 (most acidic) to 14 (most basic), with 7 as neutral. The corrosive behaviour of Lake Magadi water on flamingo legs and the pitting of the aluminium sufuria by lemon juice are both explained by excess ions (OH^- and H^+ respectively) reacting with biological or metallic surfaces. Salts such as NaCl can be neither acidic nor basic — a finding that raises new questions about salt classification, to be explored from Lesson 3 onwards.

LESSON 3 (80 minutes (2 × 40-minute lessons)): Lesson 3: Biological and Chemical Roles of Acids and Bases — Digestion, Respiration and Bleaching	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Students explain the biological and chemical roles of acids and bases — in digestion, blood respiration, and bleaching — and connect these roles to the anchoring phenomenon: why the caustic alkaline water of Lake Magadi can support flamingo life, and why the acidic lemon juice that attacked the aluminium sufuria overnight is safe to drink and vital for digestion.
Knowledge	Students will know that: (i) the stomach is an acidic environment (hydrochloric acid, HCl, pH 1–2) that activates enzymes and kills pathogens; (ii) the small intestine requires a basic environment (pH 7–8, supplied by pancreatic sodium hydrogen carbonate) for intestinal enzyme activity; (iii) antacids are weak bases that neutralise excess stomach acid; (iv) CO_2 produced during cellular respiration dissolves in blood plasma to form carbonic acid (H_2CO_3), maintaining blood pH around 7.4; (v) bleaching agents such as sodium hypochlorite (NaOCl) are alkaline and destroy colour pigments through oxidation.
Skills	Students will be able to: (i) construct and annotate a diagram of the human digestive tract marking pH zones; (ii) use evidence from readings and peer discussion to explain why neutralisation is essential at the duodenum; (iii) write and interpret the equation $CO_2 + H_2O \rightleftharpoons H_2CO_3$ in the context of respiration; (iv) revise their cumulative model diagram to incorporate biological acid/base roles; (v) pose and record refined driving questions on the Driving Question Board.
Attitudes	Students will appreciate that acids and bases are not simply 'dangerous chemicals' but are precisely regulated agents essential for life — reflected in the careful pH balance of their own bodies, in the flamingos surviving Lake Magadi's caustic water, and in communities safely using wood-ash lye and lemon juice for centuries.
Key Inquiry Question	1. Why are salts important in day-to-day life? 2. How do salts behave when exposed to air?
Purpose in Storyline	Lesson 3 is the EXPLAIN phase of the 8-lesson evidence-gathering sequence. Lessons 1–2 established what acids, bases, and salts are and how to measure pH. Lesson 3 uses the antacid supporting phenomenon to deepen students' understanding of WHY biological systems need both acids and bases, and connects this back to the anchoring phenomenon: the flamingo gut must neutralise Lake Magadi's alkaline water during digestion; the same chemistry governs the lemon juice that pitted the sufuria. This lesson supplies the 'biological role' layer students need before Lesson 4 (salt classification and hygroscopy) and Lesson 5 (neutralisation reactions).
Safety Notes	No concentrated acids or bases are handled in this lesson. If dilute HCl or NaOH are demonstrated by the teacher, safety goggles and gloves must be worn and a water source must be within reach. Sodium hypochlorite (bleach) solutions should be prepared by the teacher only, kept in sealed containers, and used only in a ventilated space; learners must not inhale fumes. Antacid tablets are safe to handle but must not be consumed during the lesson. Dispose of all solutions down the drain with copious water.

B. LESSON OVERVIEW

This 80-minute lesson is the EXPLAIN phase of the acids, bases and salts unit. Using the anchoring phenomenon of Lake Magadi's caustic soda crusts and the lemon-juice-pitted aluminium sufuria as a conceptual anchor, students investigate three biological and chemical contexts in which acid–base chemistry is essential: (1) gastric digestion — the stomach's HCl environment and the neutralisation at the duodenum by pancreatic $NaHCO_3$; (2) blood respiration — $CO_2 + H_2O \rightleftharpoons H_2CO_3$ and blood pH regulation; and (3) bleaching action — sodium hypochlorite as an alkaline oxidising agent. A teacher-led antacid demonstration (Predict–Observe–Explain mini-sequence) opens the lesson. Structured peer-discussion readings, annotated diagram construction, and a model-revision activity form the core learning experience. The lesson closes with a Driving Question Board update and a formative exit card.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Phase 1 — PREDICT (Antacid Phenomenon Hook) 📹 **VIDEO:** Video: Neuron Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Properties of Acids (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Students view the two anchoring phenomenon images again (Lake Magadi shoreline; lemon-juice-pitted sufuria) displayed on screen or printed cards. Teacher holds up a tablet of Maalox or Gaviscon (a common antacid available at any Nairobi duka) and a small transparent cup of dilute HCl (coloured red with universal indicator to show pH ~1–2). Students are asked: 'This antacid tablet is a base. The red liquid represents the acid environment inside your stomach right now. Predict: what will happen to the colour of the liquid when I drop the tablet in? What is happening at a particle level? Write your prediction in your notebook — two sentences minimum.' Students write individually for 3 minutes, then share with a shoulder partner for 1 minute.	Display both phenomenon images and say: 'Before we read anything, look again at our two mysteries. The water at Lake Magadi is strongly alkaline — pH around 10. Yet flamingos wade in it and filter-feed all day. How does a flamingo's gut survive that alkalinity? And why did lemon juice — something you squeeze on your nyama choma — destroy a metal pot overnight yet never hurt your mouth?' Allow 30 seconds of silent thinking. Then hold up the antacid tablet and the indicator-coloured HCl cup. Say: 'I want you to predict, in writing, what happens when I drop this tablet into this acid — and WHY.' Circulate for 3 minutes reading predictions. Cold-call 3 students using a random name stick: ask each to read their prediction aloud. After each, say: 'Thank you — hold that prediction. We will test it in a moment.' Do NOT confirm or correct yet. WAIT TIME after each cold-call: 10 seconds of silence before moving on.	Prediction Writing with Phenomenon Anchoring — students activate prior knowledge (Lessons 1–2 pH scale, acid = H^+ ions, base = OH^- ions) and generate a testable claim before observation. This surfaces misconceptions (e.g. 'the tablet will make the liquid more acidic') that the teacher can target during the Explain phase.	Teacher reads over shoulders during the 3-minute writing period. Observable indicator: students should reference pH change, H^+ or OH^- ions, or the word 'neutralisation' from Lesson 2. Students who write only 'it will fizz' or 'it will change colour' without a mechanistic reason are flagged for targeted questioning in Phase 3.
Phase 2 — OBSERVE (Antacid Demo + Three Reading Stations) 📹 **VIDEO:** Video: Neuron Source: PhET Interactive Simulations (English) 🔍 Search ARES	PART A — Live Demonstration (8 min): Teacher drops one antacid tablet into the indicator-coloured HCl cup. Students observe the colour change (red → orange → yellow-green as pH rises toward neutral), record the colour at 30-second intervals in a simple table, and note the CO_2 effervescence. Students answer in notebooks: 'What is the evidence that a chemical change occurred? What	PART A: Drop the antacid tablet and say: 'Watch carefully — record the colour every 30 seconds in your table. Do not speak yet — just observe.' Maintain silence for 2 minutes. After 2 minutes say: 'Now write: two observations you made, and one question this raises for you.' Cold-call 2 students to share their question — write both questions on the board under 'New questions from observation.' PART	Carousel Reading with Focused Note-Taking — each card is deliberately designed to give students one 'anchor point' connecting the biological role back to the phenomenon (e.g. Card 1 ends: 'The flamingo's stomach also produces HCl; its intestine, like yours, needs a neutralisation step — powered by the same chemistry as the Lake Magadi soda'). Focused questions prevent	Teacher checks notebooks during rotation: (i) Can students write the $CO_2 + H_2O$ equation from Card 2 without copying word-for-word — i.e. do they understand the arrow direction? (ii) Can students identify the reagent that neutralises stomach acid at the duodenum? (iii) Do students connect bleach pH (~11–13) to Lake Magadi pH (~10.5) unprompted? Students who cannot locate the pH value on

for similar videos 📖 **HTML:** [Properties of Acids (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	does the colour tell you about the pH?' PART B — Reading Stations (20 min): Students rotate through three printed reading cards (one per group of 4, 8 minutes each, with a 1-minute transition bell). Each card contains a short text (~200 words), a labelled diagram, and 2 focus questions. Card 1 — 'The Stomach and the Duodenum': describes HCl production in the stomach (pH 1–2), pepsin activation, and the role of pancreatic $NaHCO_3$ in raising pH to 7–8 at the duodenum; includes a diagram of the alimentary canal with pH values labelled at each region. Card 2 — 'CO_2 in Your Blood: The Acid That Keeps You Alive': describes the reversible equation $CO_2 + H_2O \rightleftharpoons H_2CO_3 \rightleftharpoons H^+ + HCO_3^-$, explains how blood pH is maintained at ~7.4, and references how athletes like Eliud Kipchoge and Faith Kipyegon manage CO_2 during intense exercise at high altitude in the Rift Valley. Card 3 — 'Bleach in the Laundry and the Lake': describes sodium hypochlorite (NaOCl, pH ~11–13) as an alkaline oxidising bleaching agent, its use in Kenyan hospitals and laundries, and compares its pH to Lake Magadi water (pH ~10.5).	B: Before rotating, give explicit instructions: 'You have 8 minutes per card. Read the text, study the diagram, and answer the two questions in your notebook. You will NOT share answers now — you will use them in the next phase. If you finish early, re-read and underline the most surprising sentence.' Set a visible timer. During rotation, circulate and prompt with: 'Can you find where the pH value is mentioned?' and 'How does this connect to our sufuria or Lake Magadi?' — at least one targeted prompt per group per rotation. WAIT TIME after every prompt: 12 seconds minimum before speaking again.	passive reading and prime students for the Explain phase jigsaw.	Card 1's diagram are paired with a stronger reader during Phase 3.
Phase 3 — EXPLAIN (Jigsaw Expert-Share + Annotated Diagram Construction) 📹 **VIDEO:** [Video: Neuron] Source: PhET Interactive Simulations	PART A — Jigsaw Expert Share (12 min): Groups are reshuffled so that each new group of 3 contains one 'expert' from each reading card. Each expert has 3 minutes to teach their card's key idea to their new group, using their notebook as reference. The group then collaborates to answer a shared synthesis question posted on the board: 'Lake Magadi water has a pH of about 10.5. A flamingo	Before the jigsaw: 'You are now an expert on your card. Your job is to teach — not just tell. Use a specific example or number from your card. You have 3 minutes each.' During jigsaw, circulate and listen for misconceptions: if a student says 'the stomach is basic' correct immediately by asking: 'What colour did our indicator show at the start of the demo — and what pH does that correspond to?'	Jigsaw Expert Teaching + Annotated Scientific Diagram — jigsaw ensures every student is accountable for one body of evidence and must communicate it clearly (NGSS Science and Engineering Practice 8: Obtaining, Evaluating, and Communicating Information). The annotated diagram is a representational sensemaking tool (NGSS SEP 2: Developing and Using Models)	Collect 4–5 annotated diagrams (selected from different ability levels) during the transition to Phase 4. Check: (i) Are pH values at each gut region correct? (ii) Is $NaHCO_3$ (or 'sodium hydrogen carbonate') named at the duodenum? (iii) Is the $CO_2 \rightarrow H_2CO_3$ equation present and correctly placed? (iv) Does the phenomenon-connection sentence go beyond 'both are chemicals' to

(English) 🔍 Search ARES for similar videos 📖 **HTML:** Properties of Acids (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	drinks this water and processes it through a digestive system very similar to yours. Use ALL THREE reading cards to explain: (a) how the flamingo's stomach handles this alkaline intake, (b) what happens to CO_2 produced during the flamingo's intense wading activity, and (c) why the same alkalinity that attacks the sufuria is actually what makes Lake Magadi's soda commercially valuable as a bleaching/cleaning agent.' Groups write a joint paragraph answer (5 min). PART B — Annotated Diagram (10 min): Each student individually draws and labels a diagram of the human digestive system (or copies the outline provided on the resource card) and annotates it with: (i) the name and formula of the acid in the stomach; (ii) the pH range at each labelled region (mouth, stomach, duodenum, small intestine, large intestine); (iii) the name and formula of the base that neutralises acid at the duodenum; (iv) a small arrow labelled '$CO_2 \rightarrow H_2CO_3 \rightarrow$ blood buffer' pointing from the intestinal wall to a stylised blood vessel. Students also add one sentence at the bottom: 'This connects to Lake Magadi because ___.'	After jigsaw, pose the synthesis question verbally AND write it on the board. Say: 'You have 5 minutes. Every person in the group must contribute at least one sentence to the paragraph. I will cold-call one person per group to read.' Cold-call 3 groups (rotate to groups not yet called this lesson). After each reading, respond with: 'What is one thing that group said that surprised you or that you would add to?' — WAIT TIME 12 seconds. For the annotated diagram: 'This is individual work. Silence for 10 minutes. Your diagram is evidence of YOUR understanding.' Circulate and use targeted questions: 'Why does the pH go UP at the duodenum — what is added?' and 'Where in this diagram does the CO_2 you produce when you run from Kibera to school actually affect your body chemistry?'	that forces students to spatially organise pH values and chemical identities, making the abstract concrete. The phenomenon connection sentence at the bottom of the diagram ensures every student explicitly links biological chemistry to the anchoring context.	reference alkalinity, pH, or a specific reaction type? Diagrams lacking (i) or (ii) indicate the student needs re-teaching in the next lesson's opening review.
Phase 4 — DRIVING QUESTION BOARD (DQB) Update 📹 **VIDEO:** Video: Neuron Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos	Each student receives two sticky notes (or writes in the DQB section of their notebook if physical notes are unavailable). On the YELLOW note they write: 'One thing I can now explain about our Lake Magadi / sufuria mystery that I could NOT explain before Lesson 3.' On the BLUE note they write: 'One new question this lesson has raised for me.' Students post/record their notes. The class	Say: 'Our Driving Question asks why some substances from our Kenyan environment react so differently with metals, food, and living things. Look at your annotated diagram. What layer of the answer can you now fill in that was blank after Lesson 2?' Give 2 minutes of silent thinking. Distribute sticky notes. After posting, lead the gallery scan: 'Walk slowly — read, tick, think. No	Two-Colour DQB Protocol — the yellow/blue distinction separates consolidated understanding from productive uncertainty, making students' epistemic progress visible to themselves and the teacher. The gallery scan with ticking creates a low-stakes peer-review moment and surfaces consensus versus disagreement, driving further inquiry (NGSS SEP 1: Asking Questions and Defining	Teacher photographs or reads all yellow notes (or scans the notebook DQB section) after class. Indicator of success: at least 70% of yellow notes reference a specific chemical agent (HCl, $NaHCO_3$, NaOCl, H_2CO_3) or a specific pH value, rather than vague statements like 'acids are in the stomach.' Blue notes that ask about salt behaviour in air ('why does the soda crust form at Lake

📖 **HTML:** [Properties of Acids (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	then does a 3-minute 'gallery scan' — reading 3–4 peers' yellow notes and placing a tick next to any yellow note that matches their own new understanding. Teacher reads 3 yellow notes and 3 blue notes aloud, pausing after each to ask: 'Does anyone disagree with this explanation?' or 'Does anyone think they can already answer this new question?'	talking.' After 3 minutes: 'Return to your seat.' Read 3 yellow notes aloud. After each: 'Thumbs up if you would add this to YOUR explanation of the Lake Magadi mystery. Thumbs sideways if you're not sure. Thumbs down if you disagree.' Tally visible responses and say: 'I see some thumbs sideways on this one — we'll come back to that in Lesson 4 when we look at how salts behave in air.' Read 3 blue notes and write the two most scientifically generative questions on a dedicated 'Growing Questions' section of the class DQB (or board). Say: 'These two questions — [read them] — are exactly what Lessons 4 and 5 will help us answer.'	Problems).	Magadi and not dissolve?') indicate readiness for Lesson 4 content.
Phase 5 — MODEL BUILDING (Cumulative Model Revision) 📹 **VIDEO:** Video: Neuron Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Properties of Acids (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Students retrieve their cumulative model diagram begun in Lesson 1 (a two-panel sketch showing the Lake Magadi shoreline and the sufuria). They now add a third panel: a simplified cross-section of a human body showing the stomach, duodenum, and a blood vessel, with colour-coded pH zones (red = acidic, blue = basic, green = neutral). They draw arrows showing: (1) HCl produced in stomach → low pH environment; (2) $NaHCO_3$ from pancreas → neutralisation → pH rises at duodenum; (3) CO_2 from respiring cells → blood → H_2CO_3 → controlled pH 7.4. Below the three-panel model they write a 'model caption' of exactly 3 sentences that links all three panels: what the sufuria, the lake, and the human body have in common chemically. Students who finish early add a fourth annotation: 'How does a flamingo's digestive chemistry	Distribute or ask students to open their Lesson 1 model. Say: 'Your model has grown. It started with two images — a lake and a pot. Today you discovered that the same acid-base chemistry runs inside every living body right now, including yours. Add the third panel. You have 10 minutes. Your model must show at least three labelled pH values, two named chemicals, and one reaction arrow with a word equation beside it.' Circulate and prompt: 'Which panel connects to the soda crust at Magadi? Which connects to the lemon juice on the sufuria? Which is new today?' With 3 minutes remaining, say: 'Start your 3-sentence caption. Sentence 1: what the sufuria mystery and your stomach have in common. Sentence 2: what Lake Magadi water and your blood have in common. Sentence 3: what question your model still cannot	Cumulative Model Revision (NGSS SEP 2: Developing and Using Models) — the three-panel model is a cross-lesson artifact that makes learning progression tangible. Requiring a minimum of three pH values, two named chemicals, and one word equation sets a concrete cognitive floor. The 3-sentence caption forces integrative writing that connects all evidence gathered so far, consolidating the lesson's sensemaking and previewing the next lesson's focus on salt behaviour.	Spot-check 6 models (2 high, 2 mid, 2 emerging students) during the drawing period. Success criteria: (i) Three pH values present and correctly placed (stomach ~1–2, duodenum ~7–8, blood ~7.4); (ii) HCl and $NaHCO_3$ named in the correct panels; (iii) $CO_2 + H_2O \rightarrow H_2CO_3$ shown with an arrow; (iv) Caption sentence 1 makes a mechanistic comparison (not just 'both involve acids'). Models missing (iii) indicate the CO_2/blood buffer concept needs reinforcement at the start of Lesson 4. Collect 4 models as a formative sample to inform lesson 4 planning.

	compare to the human model I have drawn?'	answer.' Cold-call 2 students to read their caption sentence 3 aloud. Write both unanswered questions on the board and say: 'These are your targets for Lesson 4.'		

D. TEACHER REFLECTION

After the lesson, reflect on the following questions and record brief notes in your professional journal:

1. PREDICTION QUALITY: Were students' predictions in Phase 1 mechanistic (referencing ions, pH, or neutralisation) or purely observational (colour change, fizzing)? If mostly observational, plan a 5-minute opening review at the start of Lesson 4 revisiting the H^+/OH^- ion model from Lesson 2.

2. READING ENGAGEMENT: Which of the three reading cards generated the most discussion and which the least? If Card 2 (CO_2/blood) was least engaged, consider whether the Eliud Kipchoge/Faith Kipyegon context was sufficiently foregrounded — consider adding a short 60-second video clip of a Kenyan athlete at altitude for Lesson 4's opener.

3. JIGSAW EQUITY: Did all three 'expert' roles receive equal airtime during the jigsaw share? Were female students as likely to be appointed expert as male students? Note any gender imbalance for deliberate cold-calling adjustments in Lesson 4.

4. MODEL DEPTH: How many students' caption sentences moved beyond surface description ('acids are dangerous') to mechanistic explanation ('HCl provides H^+ ions that activate pepsin, but the same H^+ ions corrode aluminium — the sufuria — because aluminium reacts with H^+ to release hydrogen gas')? This ratio tells you how ready the class is for Lesson 4's quantitative work on neutralisation.

5. PHENOMENON CONNECTION: Could every student in the room articulate — in their own words — at least one link between the biological content of today's lesson and the Lake Magadi / sufuria anchoring phenomenon? If not, which phase broke down? Use this to refine the pacing of Phase 3's synthesis question in future iterations.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Students observed: (1) an antacid tablet neutralising indicator-coloured dilute HCl in real time, watching the pH rise from ~1–2 (red) toward neutral (yellow-green) with visible CO_2 effervescence; (2) a labelled diagram on Reading Card 1 showing pH values at each region of the human digestive tract; (3) the reversible equation $CO_2 + H_2O \rightleftharpoons H_2CO_3$ on Reading Card 2 in the context of Kenyan long-distance athletes and high-altitude running in the Rift Valley; (4) a comparison on Reading Card 3 between the pH of sodium hypochlorite bleach (~11–13) and Lake Magadi water (~10.5).
What did I learn?	Students learned that: (1) the stomach produces HCl (pH 1–2), which activates digestive enzymes and kills pathogens in food such as nyama choma or githeri; (2) the pancreas secretes sodium hydrogen carbonate ($NaHCO_3$) into the duodenum to neutralise stomach acid, raising the pH to 7–8 for intestinal enzyme function; (3) CO_2 produced during cellular respiration dissolves in blood to form carbonic acid (H_2CO_3), and the blood buffer system maintains pH at ~7.4; (4) sodium hypochlorite (NaOCl) is an alkaline bleaching agent that destroys colour pigments by oxidation — its high pH is comparable to Lake Magadi's caustic water; (5) acids and bases are not simply hazardous chemicals but are precisely regulated agents essential for life processes.
How does this explain the phenomenon?	Students explained: (1) why a flamingo wading in Lake Magadi's alkaline water (pH ~10.5) can survive — its digestive system, like the human system, uses HCl to create an acidic stomach environment, and the alkaline lake water is neutralised during digestion; (2) why the lemon juice (citric acid) that is safe to drink can simultaneously pit an aluminium sufuria overnight — the H^+ ions from the acid react with the aluminium metal (as studied in Lessons 1–2), a reaction that does not occur inside the body because the stomach lining is protected by mucus; (3) why blood must maintain a near-neutral pH of 7.4 — even though CO_2 from respiration continuously generates carbonic acid, the bicarbonate buffer system ($NaHCO_3/H_2CO_3$) prevents dangerous acidification; (4) how

	the soda ash (Na_2CO_3) harvested commercially from Lake Magadi functions as an alkaline industrial bleaching and cleaning agent — the same category of chemistry as the sodium hypochlorite used in Kenyan laundries and hospitals.

LESSON 4 (40 minutes): Reactions of Acids with Bases, Carbonates and Metals — Evidence from the Sufuria and the Antacid	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Students gather experimental evidence for how acids react with bases, carbonates and metals — directly explaining the pitted sufuria and the antacid phenomenon — and write word and balanced chemical equations to represent those reactions.
Knowledge	Students will be able to: (1) state that neutralisation is the reaction between an acid and a base producing a salt and water; (2) write word and balanced chemical equations for HCl + NaOH, H2SO4 + CaCO3, and HCl + Mg; (3) identify the gas products of acid-carbonate reactions (CO2) and acid-metal reactions (H2); (4) name the salt produced in each reaction; (5) link each reaction type back to the anchoring phenomenon.
Skills	Careful handling of chemicals using standard safety procedures; accurate observation and recording of colour changes, gas evolution and temperature; writing and balancing chemical equations; plenary presentation of experimental findings; peer questioning and critique.
Attitudes	Curiosity about why household chemicals react; responsibility and discipline in laboratory safety; appreciation that chemical reactions — even destructive ones such as the pitting of the sufuria — follow predictable patterns that can be understood and managed.
Key Inquiry Question	Why are salts important in day-to-day life? This lesson shows that salts are always the product when acids react with bases, carbonates or metals — making salt formation a central chemical event linking the Lake Magadi trona, antacids, and mineral scale in cooking pots.
Purpose in Storyline	This is the OBSERVE and INVESTIGATE lesson. Students move from describing acids and bases (Lessons 1-3) to watching them react and producing evidence. The three experiments provide the chemical foundation needed to explain the pitted sufuria (acid-metal), the antacid action (neutralisation), and the mineral crusts at Lake Magadi (carbonate chemistry). Results feed directly into Lesson 5 on salts and Lesson 7 model building.
Safety Notes	

B. LESSON OVERVIEW
In 40 minutes students carry out three short, supervised experiments — (i) acid-base neutralisation monitored with universal indicator, (ii) dilute sulfuric acid reacting with marble chips or crushed eggshell as a carbonate proxy for antacid chemistry, and (iii) dilute hydrochloric acid reacting with magnesium ribbon to model the pitting of the aluminium sufuria. Between and after experiments students write word equations, attempt balanced equations with teacher scaffolding, and share findings in a brief plenary. The lesson closes with DQB updates and setup of the petri-dish mass-change investigation for Lesson 5.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** [Experimental probability] Source: Khan	Students sit in their pre-assigned groups of four. The teacher projects both anchoring phenomenon images — Lake Magadi and the pitted sufuria —	Teacher says: Look carefully at the inside of this sufuria. The owner told us she left dilute lemon juice in it overnight. By morning, the metal surface looked dull and pitted. I	Activation of prior knowledge from Lessons 1-3 (acids contain H+ ions; lemon juice pH 2-3; antacids are bases). Predictions create cognitive dissonance that	Teacher reads sticky notes during the 60-second group share to gauge whether students connect acidity with reactivity toward metals and whether they have any

Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Calculating pH of Salt Solutions (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	side by side. Each student silently reads two prompt questions on the board for 60 seconds: (a) What do you think happens to the metal of the sufuria when lemon juice sits in it overnight — does the metal change, and if so, where does it go? (b) When you swallow an antacid tablet, it fizzes in your stomach — what do you think is being produced? Students write individual predictions in their science notebooks (2 minutes). Then groups share predictions aloud for 1 minute. Teacher collects one sentence per group on a sticky note and posts them under the DQB column headed Lesson 4 Predictions. Expected predictions include: the metal dissolves into the juice; a gas is made; the tablet neutralises the acid; heat is released. These are not corrected at this stage.	want you to think about what the lemon juice — which we now know is an acid — could be doing to the aluminium metal. Write your best scientific prediction. You have 2 minutes, working alone first. [WAIT 2 minutes in silence, circulating to read notebooks without commenting.] Teacher then says: Share your prediction with your group and agree on one sentence. You have 60 seconds. [WAIT 60 seconds.] Teacher collects sticky notes, reads two aloud without judgment, and posts them. Teacher says: By the end of today we will have experimental evidence to test every one of these predictions. Let us also set up our thinking about the antacid fizzing — that fizzing is a clue we are going to chase.	experiments will resolve. Sticky notes make predictions public and accountable.	prior model of gas production. This informs how much scaffolding to provide during Experiment 3.
Observe Phase 📹 **VIDEO:** Experimental probability Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Calculating pH of Salt Solutions (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Students conduct three sequential experiments, each lasting approximately 6-7 minutes including setup and clean-up. The teacher announces each experiment with a brief context link before starting. EXPERIMENT 1 — Neutralisation (acid + base): Each group receives a test tube with 5 mL of 1 mol/L NaOH solution already coloured blue-green by universal indicator. Students use a dropper to add 1 mol/L HCl drop by drop, gently swirling after each drop, and record colour at every 3 drops in a results table. They observe the indicator colour moving from blue-green through green to yellow and finally red as excess acid is added. Students record the pH range associated with each colour using their pH chart from Lesson 2. They	Before Experiment 1 teacher says: This first experiment models what happens when an antacid tablet meets stomach acid. The universal indicator is our witness — it will record the pH story as we add acid to the base. Watch the colour sequence carefully and connect it to the pH scale we built in Lesson 2. [WAIT for students to begin, circulate, prompt: What colour is it now? What pH does that suggest?] Before Experiment 2 teacher says: Marble chips are mainly calcium carbonate — the same family of compound as the trona crust at Lake Magadi, which contains sodium carbonate. Eggshell is also calcium carbonate — the same material your antacid tablet may contain. Watch what happens when an acid meets a carbonate. [WAIT. After fizzing	Each experiment is framed with a phenomenon context before it begins, so students are observing with purpose rather than following steps blindly. The results table structure — reactants, observations, colour change, gas test, pH change — requires interpretive language, not just description. Connecting the squeaky pop and the damp litmus test to named gases bridges observations to chemical identity.	Teacher checks results tables during circulation for: completeness of observations; correct gas test interpretation (squeaky pop = H2; damp litmus reddening at flask mouth = CO2); whether students note temperature change in Experiment 1. Groups whose tables are incomplete are prompted with: What else could you observe — smell, temperature, appearance of the solid?

	note any temperature change by touching the outside of the test tube carefully. EXPERIMENT 2 — Acid + carbonate: Each group receives a conical flask with 10 mL of 0.5 mol/L H_2SO_4 and 3-4 marble chips (or a teaspoon of crushed dry eggshell as a locally available carbonate). Students observe vigorous fizzing, collect the gas by holding damp blue litmus paper at the mouth of the flask and observe it turning red (confirming CO_2, which forms carbonic acid with water). They also observe the marble chips becoming smaller. Students record observations in their table. EXPERIMENT 3 — Acid + metal: Each group receives a test tube with 5 mL of 1 mol/L HCl and one 2 cm strip of magnesium ribbon. Teacher explicitly says no flames before students begin. Students drop the magnesium ribbon in and observe vigorous fizzing and heat production. They hold a glowing splint (pre-prepared by teacher) at the mouth of the test tube and observe the squeaky pop confirming hydrogen gas. They note the magnesium ribbon disappearing and the solution remaining colourless. All observations go into the results table with columns: reactants, observations, colour change, gas test, estimated pH change.	begins, ask: Where is this gas coming from? What is happening to the marble?] Before Experiment 3 teacher says: This experiment models the lemon juice in the sufuria overnight. We are using magnesium instead of aluminium because it reacts more visibly and safely at this concentration, but the chemistry is the same family. No flames. I will bring the glowing splint to each group. [WAIT. After squeaky pop, say: That sound is your evidence. What gas makes a squeaky pop with a glowing splint?] Teacher circulates continuously, asking probing questions rather than providing answers.		
Explain Phase 📹 **VIDEO:** [Experimental probability] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos	After experiments are cleared and benches wiped, students remain in groups for the explanation phase (approximately 10 minutes). Teacher projects a three-column scaffold on the board: Reaction Type | Word Equation | Balanced Equation. Students first complete the word equations independently	Teacher says: Before you write anything, I want you to look at your three sets of observations and tell me: in every case, what new substances were formed? [WAIT 30 seconds.] Teacher takes two responses then says: In chemistry we represent that transformation with an equation. Write the word	The three-column scaffold structures the cognitive load — word equation before balanced equation. Cold-calling after group work ensures accountability. Circling the salt product in each equation and naming it in Kenyan industrial contexts (gypsum in construction, sodium chloride in	Teacher checks the three equations on the board for: correct products named; atoms balanced on both sides; salt identified and named. Students who cannot name the salt product are asked: What is the metal ion from the base or metal? What is the negative ion from the acid?

📖 **HTML:** [Calculating pH of Salt Solutions (Flexbook)] Source: CK-12 🔍 [Search ARES for similar readings]	in their notebooks (3 minutes), then share with their group and attempt to balance the equations (4 minutes). The three equations are: (1) hydrochloric acid + sodium hydroxide produces sodium chloride + water; (2) sulfuric acid + calcium carbonate produces calcium sulfate + water + carbon dioxide; (3) hydrochloric acid + magnesium produces magnesium chloride + hydrogen. Teacher then cold-calls one group per equation to write their version on the board. Whole class checks and corrects together. Teacher introduces the general pattern: acid + base gives salt + water (neutralisation); acid + carbonate gives salt + water + carbon dioxide; acid + metal gives salt + hydrogen. Students annotate their notebook equations by circling the salt product in each reaction and labelling it. Teacher explicitly connects: The salt in Experiment 1 is sodium chloride — ordinary cooking salt. The salt in Experiment 2 is calcium sulfate — gypsum, used in construction. The salt in Experiment 3 is magnesium chloride — used in food processing and medicine. Every reaction involving an acid produces a salt. Lake Magadi is full of sodium carbonate — a salt. The mineral scale in the sufuria is a salt. Salts are everywhere and they are always products of these reactions.	equation for Experiment 1 first — acid plus base. You have 3 minutes, alone. [WAIT 3 minutes without interrupting.] Teacher then says: Compare your word equation with your group. If you disagree, decide together who is right and why. Then try to balance the equation using the atomic symbols. You have 4 minutes. [WAIT 4 minutes.] Teacher cold-calls: Group 3, send one person to write your Experiment 1 equation on the board. Everyone else, read it and decide silently whether you agree before anyone speaks. [WAIT 20 seconds after equation is written.] Teacher facilitates whole-class correction calmly, asking: Does the number of hydrogen atoms balance on both sides? Count with me.	cooking) connects abstract equations to material reality. The explicit generalisation — every acid reaction produces a salt — plants the conceptual seed for Lesson 5.	Combine them — that is your salt. Exit check: teacher asks all students to write the name of one salt produced today and say one place in Kenya where that salt or a related salt is found.
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Experimental probability] Source: Khan	With 5 minutes remaining after the Explain Phase, the class stands at the DQB. Teacher reads aloud the sticky-note predictions posted at the start of this lesson. Students vote by show of hands on which predictions today's experiments have now answered. Answered	Teacher says: Look at what you predicted at the start of this lesson. Hold up green cards if you think today has given you evidence to answer prediction number one — that the metal changes when acid touches it. [WAIT 10 seconds for card show.] Green dot goes on.	Green-dotting answered predictions makes progress visible and builds confidence. New-question generation keeps the inquiry open and student-owned. Publicly linking a new question to Lesson 5 creates anticipation and narrative continuity across the	Teacher notes which groups generate genuine mechanistic questions (why, how, what if) versus procedural ones, as an indicator of inquiry depth. The petri-dish setup requires correct use of the balance — teacher checks each group masses the

Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Calculating pH of Salt Solutions (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	predictions get a green dot sticker. Then each group writes one new question that arose from today's experiments on a fresh sticky note and posts it in the New Questions column. Expected new questions include: Why does the sufuria pit but not dissolve completely? Why does the neutralisation reaction produce heat? Do all acids react with all metals? If antacids produce CO2 in the stomach, does that cause discomfort? What happens to the salt that forms in the test tube — can we get it back? Teacher reads two or three new questions aloud, affirms them as genuine scientific questions, and tells students that the question about what salts do after they are formed will drive Lesson 5. Teacher then announces: Before you leave today, each group will set up a petri dish with one of four salt samples — sodium chloride, anhydrous calcium chloride, washing soda crystals, or copper sulfate crystals — weigh them on the balance, and leave them open on the shelf. We will observe and reweigh them at the start of Lesson 5 tomorrow.	Now — what new questions did today raise? Something you did not expect, something that surprised you, or something you want to know more about. Write one question per group. You have 90 seconds. [WAIT 90 seconds.] Teacher reads two new questions: This group asks whether all metals react with all acids — excellent. That is a testable question. This group asks why heat was released in neutralisation — that is a question about energy in reactions, which connects to chemistry at a deeper level. Both go on the board. Now let us set up our Lesson 5 investigation before the bell.	sequence.	sample and records it before leaving, serving as a science-skills formative check.
Model Building Phase 📹 **VIDEO:** Experimental probability Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Calculating pH of Salt Solutions (Flexbook)	In the final 3 minutes students individually update their personal model diagram — begun in Lesson 2 and revised in Lesson 3 — by adding a new layer. Using a coloured pen different from previous additions, each student draws three reaction arrows from the acid box in their model: one pointing to a base (labelled neutralisation produces salt plus water), one pointing to a carbonate (labelled produces salt plus water plus CO2), and one pointing to a metal (labelled produces salt plus	Teacher says: Open your model diagram to the version you updated in Lesson 3. Using your coloured pen — a different colour from anything already on the page — add three reaction arrows from your acid box. You have 3 minutes. Work in silence because this is your personal record of what you now understand. [WAIT 3 minutes without speaking.] At the end teacher says: Hand your model in if your name is on the list I call — I will read your diagrams tonight and give you feedback at	Colour-coded layering of the model over multiple lessons externalises conceptual growth — students can literally see what they knew in Lesson 2, what they added in Lesson 3, and what they are adding now. Anchoring each arrow to a Kenya-specific example ensures the model remains connected to lived experience rather than becoming purely abstract.	Overnight review of collected model diagrams checks whether students have correctly identified three distinct reaction types, named the correct products for each, and chosen plausible Kenyan examples. Errors — such as writing water as a product of the acid-metal reaction — are corrected in written feedback and addressed in the Lesson 5 warm-up.

Source: CK-12 🔍 Search ARES for similar readings	H2). Students annotate each arrow with one real-world example from Kenya: (1) antacid in stomach, (2) acid rain on limestone buildings along the Mombasa coast, (3) lemon juice on the aluminium sufuria. Students do not share models today — this is a quick personal update. Teacher collects models from two or three students to review overnight and return with written feedback comments at the start of Lesson 5.	the start of Lesson 5. Everyone else, keep yours safe. Models that are incomplete will be caught here and supported before Lesson 5 group work.		

D. TEACHER REFLECTION
After the lesson consider: (1) Did all three experiments produce clear, observable results within the time available — if Experiment 3 was rushed, plan to allow a full minute of observation time before the splint test next time. (2) Were students able to write balanced equations independently or did most need step-by-step teacher support — if support was heavy, plan a brief equation-balancing practice task as homework before Lesson 5. (3) Did the DQB new questions show genuine curiosity about salts and their behaviour, or were questions mostly procedural — if procedural, introduce the Lesson 5 canteen-cook salt-clumping phenomenon earlier to spark interest. (4) Were safety protocols followed consistently — note any groups that needed reminders about goggles or flame proximity, and reinforce at the start of Lesson 5 before the petri-dish reweighing. (5) Review the overnight model diagrams before Lesson 5 and identify two common misconceptions to address in the warm-up.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	In Experiment 1 I observed the universal indicator colour change from blue-green to red as HCl was added to NaOH, and the test tube felt warm. In Experiment 2 I observed vigorous fizzing when H2SO4 met marble chips and damp litmus paper turned red at the mouth of the flask. In Experiment 3 I observed magnesium ribbon disappearing in HCl with rapid fizzing and a squeaky pop from the glowing splint test.
What did I learn?	I learned that acids react with bases in neutralisation to produce a salt and water; acids react with carbonates to produce a salt, water and carbon dioxide; acids react with metals to produce a salt and hydrogen gas. A salt is always one of the products when an acid reacts. The salt product depends on which acid and which base or metal react together.
How does this explain the phenomenon?	These reactions explain why the aluminium sufuria was pitted overnight — the citric acid in lemon juice reacted with the aluminium metal to form a salt and hydrogen gas, corroding the surface. They also explain why antacid tablets fizz in the stomach — the carbonate in the tablet reacts with stomach HCl to form a salt, water and CO2. The mineral crusts at Lake Magadi are salts, and salts are always formed when acids react with other substances in our environment.

LESSON 5 (40 minutes): Salts — Classification and Behaviour in Air: Why Does the Canteen Salt Clump in Mombasa Rains?	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Students will investigate and explain how different salts behave when exposed to air, classifying them as hygroscopic, deliquescent, efflorescent or neutral, and connect these behaviours to the ionic structure of salts and to real Kenyan contexts including Lake Magadi soda crusts and clumping canteen salt.
Knowledge	Students will be able to: (1) define hygroscopy, deliquescence and efflorescence with reference to water of crystallisation and atmospheric moisture; (2) classify NaCl, anhydrous CaCl2, Na2CO3.10H2O and CuSO4.5H2O according to their observed behaviour in air; (3) explain why salts differ in their affinity for atmospheric water using a model of ionic attraction; (4) write the formula of a hydrated salt and identify the water of crystallisation.
Skills	Students will practise: accurate measurement of mass using a balance; systematic recording of quantitative and qualitative observations over time; classification of substances based on experimental evidence; construction of explanatory models linking macroscopic observations to particle-level reasoning; safe handling of chemicals.
Attitudes	Students will demonstrate: curiosity when confronted with unexpected mass changes; patience in waiting for and recording 24-hour results; appreciation of how chemistry explains familiar Kenyan everyday puzzles; willingness to revise prior explanations in light of new quantitative evidence.
Key Inquiry Question	Why are salts important in day-to-day life? How do salts behave when exposed to air?
Purpose in Storyline	Lesson 5 advances the storyline from reactions that produce salts (Lesson 4) to the properties and behaviour of the salts formed. Students now have evidence that every acid reaction produces a salt; here they gather new evidence about what happens to those salts once formed — do they absorb moisture from Mombasa coastal air, lose water like the drying trona crusts at Lake Magadi, or remain unchanged like table salt in a sealed container? This evidence will feed directly into Lesson 6 (applications) and Lesson 7 (cumulative model building).
Safety Notes	

B. LESSON OVERVIEW
Students arrive at Lesson 5 to observe petri dishes of four salts that were set up and massed at the end of Lesson 4 (24 hours earlier). The lesson opens with the canteen cook supporting phenomenon — a vivid story of a school cook in Mombasa whose bulk salt supply clumps into a solid block every rainy season, making it impossible to pour. Students predict why this happens, then collect their 24-hour mass data and visual observations from the petri dish investigation. Through guided sensemaking they classify each salt, learn the vocabulary of hygroscopy, deliquescence and efflorescence, and build a particle-level model explaining why different ionic compounds interact differently with atmospheric water. The lesson closes by returning to both the Lake Magadi anchoring phenomenon (trona efflorescence and soda ash hydration) and the pitted sufuria, with students updating the Driving Question Board with refined explanations.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**

Predict Phase 📹 **VIDEO:** [Acid–base properties of salts] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Adding Mixed Fractions of Water (PLIX)] Source: CK-12 🔍 Search ARES for similar readings	Students read a short printed scenario card (or teacher reads aloud): A school canteen cook in Mombasa orders a 50 kg sack of table salt at the start of term. By mid-term, after weeks of coastal rain, the salt has fused into one hard block and the cook cannot get it out of the container. Meanwhile, the blue copper sulphate crystals the chemistry teacher stored near the window have turned pale and powdery. And at Lake Magadi, the white trona crusts that form on the shoreline in the dry season seem to shrink and crack in the rainy season. Students write individual predictions (2 minutes, silent): (1) What do you think is happening to the salt in Mombasa? (2) What do you think is happening to the blue crystals? (3) How might these observations connect to the Lake Magadi crusts? Students then share predictions with a partner (think-pair) and identify the key question: do all salts behave the same way when exposed to air?	Teacher displays the scenario on the board or reads it with expression, pausing after each of the three observations to let students visualise the context. Teacher says: Before you write anything, close your eyes for five seconds and picture the Mombasa canteen — a humid kitchen, bags of salt stacked by the wall, rain drumming on the iron roof. What is the air doing to that salt? WAIT TIME: 10 seconds of silent thinking before students write. After pair-share, teacher cold-calls three pairs and records their key words on the board under the heading What we predict. Teacher does NOT confirm or correct yet — only probes with: What evidence from Lesson 4 might help you make this prediction? What do you remember about how salts are formed from ionic compounds? WAIT TIME: 8 seconds after each probe. Teacher then says: In about five minutes we are going to look at the petri dishes we set up yesterday. Keep your predictions in mind — you are about to find out whether your ideas match the evidence.	Activating prior knowledge about ionic bonding and salt formation (Lesson 4 link); bridging everyday Kenyan context (Mombasa humidity, Lake Magadi) to abstract chemical behaviour; prediction-before-observation protocol to create cognitive conflict.	Teacher scans written predictions as students write — listening for whether students invoke particle-level reasoning (ions attracting water molecules) or remain at surface level (salt gets wet). This informs how deeply to probe during the Observe phase. Red flag: if most students predict nothing will happen to any salt, teacher notes this to address in Explain phase.
Observe Phase 📹 **VIDEO:** [Acid–base properties of salts] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Adding Mixed Fractions of Water (PLIX)] Source: CK-12	Groups of four collect their labelled petri dishes from the shelf. Each dish was massed and photographed (or sketched) at the end of Lesson 4. Each group has a results table already begun in their notebooks with columns: Salt name and formula | Initial mass (g) | Mass after 24 h (g) | Mass change (g) | Visual appearance at start | Visual appearance after 24 h | Classification (to fill in later). Students use the balance to re-mass each dish and record results. They also describe appearance	Teacher circulates as groups collect data, asking targeted questions: What exactly do you see — describe it as if you are writing for someone who cannot see the dish. What does the mass change number tell you about what happened? If the mass went up, what substance was added? If it went down, what was lost? WAIT TIME: 6 seconds after each question before accepting a response. Teacher facilitates class data-sharing by pointing to the board table and asking: Which salt	Quantitative data collection with mass balance grounds the discussion in measurement rather than impressions; cross-group data pooling models scientific practice of replication; teacher questioning sequence moves from observation (describe) to inference (explain the number) to connection (link to canteen story); cognitive conflict is created when $CaCl_2$ liquefies — a surprising outcome that demands explanation.	Teacher checks that all groups have recorded both quantitative (mass) and qualitative (appearance) data before moving on. Groups that have only written did not change or changed a lot are prompted to be more precise. Teacher listens for the key inference: mass increase means something was gained from the air, mass decrease means something was lost to the air — if this is not emerging naturally, teacher flags it for explicit address in Explain phase.

🔍 Search ARES for similar readings	carefully: has the salt dissolved into a liquid, absorbed moisture and become damp, turned powdery, changed colour, or shown no change? Groups then share data across the class — teacher writes a class data table on the board with each group contributing one reading per salt (four readings per salt across four groups) so students can compare and identify consistent trends despite measurement variation. Students note: anhydrous $CaCl_2$ may have partially dissolved into a liquid pool (deliquescence); $Na_2CO_3.10H_2O$ may have lost mass and turned to a white powder (efflorescence); NaCl shows minimal mass change; $CuSO_4.5H_2O$ may show slight mass loss as surface water of crystallisation releases.	showed the biggest mass increase? Which showed a mass decrease? Which showed almost no change? Teacher resists giving explanations at this stage — only asks: Is this what you predicted? What surprises you? When the $CaCl_2$ dish is discussed, teacher asks: If this salt has turned partly liquid, where did the liquid come from? WAIT TIME: 8 seconds. Teacher connects to the Mombasa canteen scenario: Could this behaviour explain what happened to the canteen salt? What do you think — is table salt the same as calcium chloride in this respect?		
Explain Phase 📹 **VIDEO:** Acid–base properties of salts Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Adding Mixed Fractions of Water (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Teacher introduces three vocabulary terms one at a time, each anchored to the data just collected. For each term, students first attempt a definition in their own words before the formal definition is given. (1) Hygroscopy: Students see that NaCl gained a tiny amount of mass — teacher asks them to define hygroscopic before revealing: a hygroscopic salt absorbs moisture from the air but does not dissolve in it. (2) Deliquescence: Students look at the $CaCl_2$ dish — teacher asks: what word would you coin for a salt that absorbs so much water it dissolves itself? Students suggest words, then receive the term deliquescent. Formal definition: a deliquescent salt absorbs enough water vapour from the air to dissolve completely into a solution. (3) Efflorescence: Students look at	Teacher uses a deliberate vocabulary-reveal sequence: student definition first, then formal definition, then immediate application to data — never giving the term before students have tried to articulate the concept. When introducing each term, teacher says: Before I tell you the word chemists use, tell me what YOU would call this behaviour based on what you just saw. WAIT TIME: 10 seconds of thinking, then pair-share, then cold call. For the particle-level diagram, teacher draws slowly on the board and asks students to narrate what they see rather than simply copying: What is happening at the surface of this crystal? What do the arrows show? WAIT TIME: 8 seconds per question. Teacher explicitly bridges to prior learning: In Lesson 4 we produced salts as products of	Vocabulary-last protocol prevents passive reception of terms; particle-level diagram makes the abstract ionic attraction model visual and annotatable; deliberate bridging to prior lessons and to both phenomena maintains the storyline coherence; Kenya-specific humidity geography (Mombasa coast vs Nairobi highlands) grounds the chemistry in lived experience.	Teacher checks classification tables — all four salts should be correctly classified. Common error to watch for: students classifying NaCl as deliquescent because it clumps in humid conditions — teacher addresses this by asking: did NaCl dissolve into a pool? No — so it is hygroscopic, not deliquescent. A brief exit-check question is posed verbally: Give me one difference between a hygroscopic salt and a deliquescent salt — students write one sentence before the next phase begins.

	Na2CO3.10H2O — it lost mass and turned powdery. Teacher asks: If the salt lost mass and the air did not take ionic material away, what must have been lost? Students deduce water. Teacher introduces efflorescence: an efflorescent salt loses water of crystallisation to the air, forming a powdery anhydrous or less-hydrated residue. Students then complete a classification table in their notebooks and write a one-sentence particle-level explanation for each behaviour, prompted by: think about how strongly the ions in each salt attract water molecules compared to how strongly the air holds onto water vapour. A short teacher-led diagram on the board shows a hydrated salt crystal with water molecules in the lattice, an anhydrous lattice pulling water from air, and an efflorescent lattice releasing water to dry air. Students copy and annotate. The class then discusses: where does CuSO4.5H2O fit — slightly efflorescent? The number 5 in its formula is linked to five water molecules per formula unit. Students revisit the Lake Magadi trona formula (Na3H(CO3)2.2H2O — sodium sesquicarbonate with water of crystallisation) and the teacher asks: In the dry Rift Valley season, what do you predict happens to the trona? In the wet season? Students connect this to the anchoring phenomenon crusts shrinking and cracking.	acid reactions. Now we are asking: once a salt is formed, what does it do next? Does it just sit there, or does it interact with its environment? Teacher connects to the sufuria phenomenon: the mineral scale inside the sufuria is partly crystallised salts from hard water — are they hygroscopic, deliquescent or efflorescent? Why does the scale feel damp sometimes? Teacher also connects to Kenyan food science: why does ugali salt need to be stored in a sealed container along the coast but less so in Nairobi (lower humidity inland)?		
Driving Question Board (DQB) Creation 📹 **VIDEO:**	Students return to the Driving Question Board posted since Lesson 1. Each group reviews the sticky notes already there and identifies which questions from	Teacher frames the DQB activity by saying: Every piece of evidence we gather should either answer an old question or open a new one — that is how science moves	DQB revisit maintains the storyline thread across all eight lessons; colour-coding answered vs new questions makes epistemic progress visible; gallery walk	Teacher reads the new ORANGE sticky notes after class to identify misconceptions or gaps that need addressing in Lesson 6. If students are not connecting salt behaviour

Acid–base properties of salts Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Adding Mixed Fractions of Water (PLIX) Source: CK-12 🔍 Search ARES for similar readings	previous lessons can now be answered using todays evidence about salt behaviour. Students write new sticky notes in two colours: GREEN for questions now answered by todays evidence; ORANGE for new questions raised by todays lesson. Examples of questions students are expected to add: Why does the trona crust at Lake Magadi look different in wet and dry seasons? Is anhydrous $CaCl_2$ used anywhere in Kenya to dry things out? If washing soda ($Na_2CO_3.10H_2O$) is efflorescent, does it lose its cleaning power over time? Why is silica gel (a desiccant) used in medicine packets — is it acting like $CaCl_2$? Groups post their sticky notes and do a brief gallery walk (2 minutes) to read other groups contributions.	forward. Look at the board — which old questions does todays petri dish evidence help you answer? WAIT TIME: 30 seconds of silent reading of the board before students write new notes. Teacher circulates during the gallery walk, pointing to specific sticky notes and asking: Is this question answered now, or do we need more evidence? Teacher ensures the anchoring phenomenon connection is made explicit by asking the whole class: Look at the original Lake Magadi image — can you now explain one more thing about it that you could not explain in Lesson 1? WAIT TIME: 8 seconds, then cold-call two students. Teacher records any consensus answers on the board under the heading What we can now explain.	creates low-stakes peer learning and reveals the range of student thinking; explicit return to the anchoring phenomenon image reinforces the unit coherence.	to applications (e.g., using desiccants, storing fertilisers), this signals that Lesson 6 applications content needs stronger scaffolding. Teacher notes whether students can articulate why the Lake Magadi trona behaves differently in wet vs dry seasons — this is a key understanding checkpoint.
Model Building Phase 📹 **VIDEO:** Acid–base properties of salts Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Adding Mixed Fractions of Water (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Students return to the model diagram they began in Lesson 2 and revised in Lesson 3. They now add a new section labelled Salt Behaviour in Air. In this section they draw and annotate: (1) a hygroscopic salt (NaCl) with small arrows showing water vapour being partially attracted but not dissolving the lattice; (2) a deliquescent salt ($CaCl_2$) with water molecules completely surrounding and separating ions; (3) an efflorescent salt ($Na_2CO_3.10H_2O$) with water molecules leaving the lattice into dry air. Below the diagram, students write a two-sentence synthesis connecting their model to the driving question: they must mention at least one Kenyan context (Lake Magadi, Mombasa canteen, hard water scale in the sufuria) and at least one chemical	Teacher says: Your model is a scientific tool — it should help you explain new situations you have not seen before. If a friend in Kisumu asked you why their baking soda goes lumpy in the lake-region humidity, could your model answer that? Try now — write it in your model. WAIT TIME: 10 seconds of thinking before writing. Teacher circulates and looks specifically for particle-level language (ions, water molecules, lattice, attraction) — if students are writing only at the surface level (salt absorbs water), teacher prompts: Can you say WHY it absorbs water — what is happening between the ions and the water molecules? WAIT TIME: 6 seconds. When selecting models to display, teacher chooses one that includes particle-level reasoning and one that does not,	Cumulative model revision across lessons makes learning progression visible and metacognitive; the test of a good model (can it predict a new situation) develops scientific reasoning; peer critique with a specific prompt (what does it explain well, what is missing) is more productive than open-ended feedback; closing with the driving question reanchors the lesson in the unit storyline.	Teacher collects model diagrams (or photographs them) at the end of the lesson to check for: correct use of hygroscopy, deliquescence and efflorescence vocabulary; presence of particle-level reasoning; connection to at least one Kenyan context; correct chemical formula for at least one hydrated salt. Students who include only surface-level descriptions will receive a written prompt before Lesson 6 asking them to add particle-level reasoning to their model.

	formula. Students share their additions with a partner and suggest one improvement to each others model. Teacher selects two contrasting models to display briefly and asks the class: What does this model explain well? What is still missing?	and asks the class: Which model would better help a scientist predict what happens to a new unknown salt in humid air? Why? WAIT TIME: 8 seconds. Teacher closes the lesson by returning to the driving question on the board and asking: What new evidence did we add today that helps us answer why some substances from our Kenyan environment react so differently with metals, food and living things? One student is cold-called to summarise in two sentences.		

D. TEACHER REFLECTION

After the lesson, consider: Did the 24-hour petri dish investigation produce clear enough results to anchor the three vocabulary terms? If the $CaCl_2$ did not fully deliquesce (possible in a dry inland school), was I prepared to use a photograph or demonstration instead? Did I successfully maintain the storyline link between the petri dish data, the Mombasa canteen salt scenario and the Lake Magadi anchoring phenomenon, or did the lesson feel like a standalone activity? Were students able to move from macroscopic observation (mass change, appearance) to particle-level explanation, or did most remain at the surface level? If particle-level reasoning was weak, plan a brief warm-up at the start of Lesson 6 where students sketch one ionic compound lattice attracting a water molecule. Did the DQB activity generate genuine new questions or were students just copying existing notes? If the DQB felt mechanical, consider assigning one group to play devil's advocate and challenge the most popular sticky note explanation. Note any students who made the Kisumu baking soda connection or spontaneously asked about desiccants — these students are ready for extension tasks in Lesson 6.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We observed four salts — NaCl, anhydrous $CaCl_2$, $Na_2CO_3.10H_2O$ and $CuSO_4.5H_2O$ — left in open petri dishes for 24 hours. We measured and recorded mass changes and changes in visual appearance. We found that $CaCl_2$ gained mass and partially dissolved into a liquid, $Na_2CO_3.10H_2O$ lost mass and turned powdery, NaCl showed very little change, and $CuSO_4.5H_2O$ showed slight mass loss.
What did I learn?	We learned that salts differ in how they interact with water vapour in the air. A hygroscopic salt absorbs moisture from the air without dissolving. A deliquescent salt absorbs so much water vapour that it dissolves completely. An efflorescent salt loses its water of crystallisation to the air and becomes a dry powder. These differences depend on how strongly the ions in the salt lattice attract water molecules compared to how strongly the atmosphere holds water vapour.
How does this explain the phenomenon?	We explained that the canteen cook's salt in Mombasa clumps because NaCl is hygroscopic — it absorbs moisture in the high coastal humidity. We explained that the Lake Magadi trona crusts can shrink and crack in the wet season because $Na_3H(CO_3)_2.2H_2O$ can release or absorb water of crystallisation depending on atmospheric humidity — a form of efflorescence or hygroscopic behaviour. We connected the water of crystallisation in hydrated salt formulas (such as $CuSO_4.5H_2O$ and $Na_2CO_3.10H_2O$) to the observable mass changes we measured, showing that the dot-number in a hydrated formula represents

	real water molecules held inside the ionic lattice that can be lost or gained.

LESSON 6 (80 minutes (2 × 40-minute lessons)): Applications of Salts in Real Life — From Lake Magadi to Your Kitchen, Farm and Clinic	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	By the end of this lesson, learners will be able to outline applications of salts in real-life situations across five sectors (agriculture, food industry, medicine, laundry, and road use) and promote proper use of salts in day-to-day life, including awareness of correct salt intake and blood pressure.
Knowledge	Learners will be able to: (i) identify at least five named salts and their specific real-life applications across agriculture (CAN, DAP), food industry (NaCl, sodium benzoate), medicine (saline, antacids — magnesium hydroxide, calcium carbonate), laundry (washing soda, $Na_2CO_3 \cdot 10H_2O$), and road gritting (NaCl); (ii) connect Lake Magadi trona/soda ash (Na_2CO_3) to industrial and domestic uses; (iii) explain why excess dietary NaCl raises blood pressure (osmotic pressure, fluid retention).
Skills	Learners will be able to: (i) research, extract and synthesise information from digital devices or print media on salt applications; (ii) construct a sector application map (graphic organiser) linking salts to uses; (iii) plan and draft a community sensitisation message on safe salt intake; (iv) deliver a structured peer-teaching segment (jigsaw) presenting one sector's findings to the class.
Attitudes	Learners will: (i) appreciate the dual nature of salts — essential yet potentially harmful — and demonstrate responsibility in promoting safe salt use in their communities; (ii) respect peers' contributions during jigsaw teaching; (iii) show patriotism by connecting Kenya's Lake Magadi soda ash industry to national economic development.
Key Inquiry Question	1. Why are salts important in day-to-day life? 2. How do salts behave when exposed to air?
Purpose in Storyline	In Lessons 1–5, learners identified acids and bases, measured pH, investigated chemical reactions of acids with metals and carbonates, and classified salts by their behaviour in air (hygroscopic, deliquescent, efflorescent). Lesson 6 is the EXPLAIN & APPLY pivot: learners now map everything they know about salts onto real-world sectors, connecting the Lake Magadi soda ash (Na_2CO_3) to the washing soda in a Nairobi household, the CAN fertiliser on a Rift Valley maize farm, and the saline drip in a Kenyatta National Hospital ward. They also take the first step in the sub-strand's community project — drafting blood-pressure awareness messages — directly addressing the anchoring phenomenon question: 'How can understanding acid, base or salt chemistry help us use these substances safely and sustainably?'
Safety Notes	No hazardous chemicals are handled in this lesson. If physical salt samples (NaCl, washing soda, CAN fertiliser granules) are passed around for observation, learners must NOT taste any substance and must wash hands after handling fertiliser granules. Fertiliser bags (CAN, DAP) carry GHS hazard pictograms — discuss these briefly as a literacy exercise. Digital devices must be used responsibly; learners follow the school's acceptable-use policy.

B. LESSON OVERVIEW

Lesson 6 is a 80-minute research-and-peer-teaching session structured around a Jigsaw cooperative learning protocol. Five expert groups each investigate one real-life sector in which salts play a central role: (A) Agriculture — CAN and DAP fertilisers on Rift Valley and Central Kenya farms; (B) Food Industry — table salt preservation of nyama choma and fish at the Coast, sodium benzoate in juice preservation; (C) Medicine — oral rehydration salts (ORS), saline drips, antacids ($Mg(OH)_2$, $CaCO_3$) for treating excess stomach acid; (D) Laundry — washing soda ($Na_2CO_3 \cdot 10H_2O$) as a water softener for githeri-pot cleaning, and its link to Lake Magadi trona; (E) Road Use / Environment — salt gritting on highland roads (Timboroa, Mau Summit), and the environmental concern of eutrophication from fertiliser run-off into Lake Victoria. Each group produces a labelled sector map, then reforms into mixed 'teaching groups' where every member teaches their sector. The lesson closes with learners

individually drafting a two-sentence community sensitisation message on safe dietary salt intake and its link to blood pressure — launching the sub-strand project. Throughout, all salt examples are explicitly reconnected to the anchoring phenomenon: the Lake Magadi soda crusts and the pitted sufuria.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Phase 1 — Predict (Activate & Connect to Phenomenon) 📹 **VIDEO:** Conservation and the race to save biodiversity Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Hydrolysis of Salts: Equations (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners view the two anchoring phenomenon images displayed on the board or projected screen: (1) Lake Magadi soda crusts with flamingos; (2) the pitted aluminium sufuria. The teacher reveals a new 'clue image': a split-screen showing (a) a bag of CAN fertiliser with a Rift Valley maize farm in the background, and (b) a hospital saline drip bag labelled '0.9% NaCl'. Learners individually write in their science notebooks for 3 minutes: 'What do the Lake Magadi trona, the CAN fertiliser bag, and the hospital saline bag have in common chemically? Where else in your daily life have you encountered a salt today?' Pairs share for 1 minute (Think-Pair-Share). Three pairs are cold-called to share with the class. Learners then vote by show of hands on which sector they think uses the most different types of salts — responses are recorded on the board as a baseline prediction to be revisited at the end.	Display images and say: 'Look carefully at these three images. You have 3 minutes — write in your notebook: what do these three things share chemically, and where did YOU encounter a salt today before you came to school?' [Start timer. Circulate silently for 90 seconds, then prompt quietly: 'Think about your breakfast — ugali, uji, mandazi — or the water you drank.'] After 3 minutes: 'Share your idea with your partner. You have 1 minute.' [Allow pairs to talk. WAIT TIME: 10 seconds after calling each pair before accepting response.] Cold-call 3 pairs using name sticks: 'Amina and Kipchoge — what did your pair decide? Thank you. Fatuma and Otieno — any different idea? Wanjiru and Hassan — do you agree or disagree, and why?' Record key ideas on board under heading: SALT CONNECTIONS. Then run the sector vote: 'Hands up if you think agriculture uses the most types of salts... food industry... medicine... laundry... roads.' Tally and note predictions visibly. Say: 'By the end of this lesson, let us see if your predictions hold.'	Think-Pair-Share with Cold-Call: activates prior knowledge from Lessons 1–5, surfaces everyday Kenyan salt encounters (dietary NaCl, water-treatment alum, fertiliser), and creates a predictive stake that motivates the evidence-gathering phase. Recording predictions on the board makes thinking visible and provides a concrete reference point for revision.	Observable indicator: Each learner has written at least one chemical commonality (e.g., 'they are all salts / ionic compounds / formed from acid + base') and one personal daily encounter in their notebook before pairs share. Teacher scans notebooks during circulation — if fewer than 60% of learners have written a chemical link, pause and prompt: 'Think back to Lesson 3 — how is a salt formed? Acid + ___?' This reveals whether the Lesson 3 definition of salts has been retained.
Phase 2 — Observe (Jigsaw Expert-Group Research) 📹 **VIDEO:**	Learners are assigned to five expert groups of ~5–6 learners (Groups A–E, one sector each). Each group receives a printed or digital Research Brief card listing: (i) the sector name and 3 focus	Distribute Research Brief cards and sector map templates. Say: 'You are now EXPERTS. Your group's job is to become the best in the class on your sector in 15 minutes. Use the Research Brief to	Jigsaw Expert Research with Structured Graphic Organiser: the sector map scaffold forces learners to organise information into the same categories across all sectors, making later peer-	Observable indicator: At the 12-minute mark, teacher checks that each group's sector map has at least 3 of 5 rows substantively completed and that the 'Link to Phenomenon' cell is not blank. If a

[Conservation and the race to save biodiversity](#) Source: Khan Academy (English - US curriculum) 🔍 [Search ARES for similar videos](#) 📖 **HTML:** [Hydrolysis of Salts: Equations (Flexbook)](#) Source: CK-12 🔍 [Search ARES for similar readings](#)	questions; (ii) a list of 2–3 specific salts to investigate; (iii) a scaffold graphic organiser (sector map template) with columns: Salt name | Chemical formula | How it is made / where it comes from | What it does in this sector | Link to Lake Magadi OR the sufuria phenomenon. Groups use available digital devices (ARES platform resources, downloaded reference sheets) or printed reference materials to research for 15 minutes. Each group must fill in their sector map AND identify at least one Kenya-specific example (e.g., Group A: CAN from Elgon Farmers, DAP imported through Mombasa Port; Group D: Tata Chemicals Magadi Ltd exports Na_2CO_3 globally). Groups also note one risk or misuse of their salt (e.g., excess NaCl → hypertension; excess fertiliser → eutrophication in Lake Victoria).	guide you. Fill in every row of your sector map. You must find at least one Kenyan example — a farm name, a brand, a place — and one risk.' [Circulate continuously, spending ~2–3 minutes per group.] For Group D (Laundry/Magadi): prompt — 'What is the chemical formula of washing soda? How does it link to what you saw at Lake Magadi in the phenomenon images?' [WAIT TIME: 12 seconds after posing prompt, before offering a hint.] For Group C (Medicine): prompt — 'An antacid tablet like Gelusil contains $Mg(OH)_2$. Is that an acid, base or salt? How does it neutralise HCl in the stomach? Write the equation.' For Group E (Roads/Environment): prompt — 'What happens when fertiliser salts from a Rift Valley farm wash into a river and reach Lake Victoria? What is that process called?' At 12 minutes: give a 3-minute warning. 'You have 3 minutes — make sure your sector map has the Kenya example and the risk row filled in.'	teaching and cross-sector comparisons efficient. Requiring a Kenya-specific example and a risk/misuse column prevents superficial listing and pushes learners toward application and evaluation (higher-order thinking). The explicit 'Link to Phenomenon' column maintains the thread back to Lake Magadi and the sufuria throughout.	group is stuck on the phenomenon link, teacher asks: 'Is your salt made by a neutralisation reaction? Could it be found at Lake Magadi or left behind when water evaporates — like the scale in the sufuria?' Groups that finish early are directed to the extension question on their Research Brief: 'Write a balanced chemical equation for one reaction that produces your sector's key salt.'
Phase 3 — Explain (Jigsaw Teaching Groups + Class Consolidation) 📹 **VIDEO:** [Conservation and the race to save biodiversity](#) Source: Khan Academy (English - US curriculum) 🔍 [Search ARES for similar videos](#) 📖 **HTML:** [Hydrolysis of](#)	Expert groups dissolve and learners reform into 5–6 mixed Teaching Groups, each containing one representative from every sector (A–E). Each expert teaches their sector to the group for 3 minutes using their completed sector map as a visual aid, totalling ~15 minutes of peer teaching. Listeners complete a Reception Table in their notebooks with columns: Sector | Key Salt(s) & Formula | Main Use | Kenya Example | Risk / Misuse | Link to Phenomenon. After all five sectors are taught, the whole class reconvenes. The teacher leads a 10-minute class consolidation	Before expert groups dissolve: 'You have 2 minutes to practise your 3-minute teach — decide which THREE most important points you will say, and make sure you can point to your sector map as you speak.' [Allow 2 minutes.] Reform into teaching groups: 'Move to your teaching group now. Expert A starts. You have exactly 3 minutes. Listeners — fill in your Reception Table as they speak.' [Use a visible timer. After each expert: 'Thank the expert. Next expert — B, your turn.'] After all five experts have taught: whole-class reconvene. Cold-call one learner per sector (5 cold-calls	Jigsaw Peer Teaching with Reception Table: the teaching role requires experts to consolidate and communicate their knowledge, deepening their own understanding (protégé effect). The Reception Table ensures listeners are active, not passive. The teacher's phenomenon-connection mini-lecture at the end provides authoritative anchoring so that any misconceptions introduced during peer teaching are corrected before the model-building phase. Revisiting the Phase 1 prediction tally makes conceptual growth visible and emotionally satisfying.	Observable indicator: After the teaching-group round, teacher spot-checks 5 Reception Tables (one per sector) — each should have at least 4 of 5 sectors' rows filled with a formula and a Kenya example. During the cold-call Master Map construction, teacher listens for: correct chemical formulae (CAN = $Ca(NO_3)_2 \cdot NH_4NO_3$ complex, but accept 'calcium ammonium nitrate'; DAP = $(NH_4)_2HPO_4$; NaCl; $Na_2CO_3 \cdot 10H_2O$; $Mg(OH)_2$ or $CaCO_3$ for antacids). If a learner gives only a trade name without a formula, teacher probes: 'Good — what is the chemical formula of

Salts: Equations (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	discussion, building a Master Sector Map on the board by calling one representative from each sector to contribute one key finding. The teacher explicitly connects the Lake Magadi Na_2CO_3 thread: 'Washing soda is $Na_2CO_3 \cdot 10H_2O$ — a hydrated form of soda ash from Lake Magadi. The same compound that forms those white crusts in our phenomenon image is in your mother's laundry bucket and in the glass factory in Nairobi Industrial Area.' The teacher also addresses the sufuria phenomenon: 'The scale inside the aluminium sufuria is calcium carbonate — $CaCO_3$ — a salt deposited from hard water. That is the same family of compounds as the antacid in Group C's sector.'	total) to contribute to the Master Sector Map: 'Akinyi — what is ONE salt and its formula from the Agriculture sector?' [WAIT TIME: 12 seconds.] 'Mwangi — one use from Medicine?' [WAIT TIME: 12 seconds.] Build the Master Map on the board systematically. Then deliver the explicit phenomenon connections (Lake Magadi washing soda link; sufuria scale $CaCO_3$ link) as a mini-explanation, pausing to ask: 'Does this change your prediction from the beginning of the lesson about which sector uses the most types of salts? Show hands again.' Revisit the tally from Phase 1.		table salt? Na... C... or Na... Cl?' Misconceptions about antacids being acids (common) are addressed immediately: 'Antacids contain bases — $Mg(OH)_2$ is a base — but the SALT it forms after neutralising stomach acid is $MgCl_2$. So both a base AND a salt are involved.'
Phase 4 — Driving Question Board (DQB) Update 📹 **VIDEO:** Conservation and the race to save biodiversity Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Hydrolysis of Salts: Equations (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners return to the class Driving Question Board. In pairs, they spend 4 minutes reviewing the sticky notes already on the board from Lessons 1–5 and do three things: (1) move any question that Lesson 6 has now answered into the 'ANSWERED' column, writing a brief evidence note on the back of the sticky note; (2) add at least one NEW question generated by today's lesson (e.g., 'If washing soda comes from Lake Magadi, does that mean Lake Magadi water is used in laundry detergent factories?'; 'Why does too much salt raise blood pressure — what is the body doing with the sodium?'; 'Can the eutrophication from fertiliser salts be reversed?'); (3) place a star sticker on any question they personally feel most curious about. The teacher draws attention to the driving question itself — written permanently at the	Direct learners to the DQB: 'Take 4 minutes with your partner. Look at the questions from previous lessons. Which ones can you now answer after today? Flip the note, write the answer evidence, and move it to ANSWERED. Add at least one brand-new question today's lesson created for you. Then put a star on the question you most want answered before the unit ends.' [Circulate and read notes being flipped — this is rapid formative data.] After 4 minutes: whole-class share — cold-call 3 pairs: 'Halima and Njoroge — which question did you move to ANSWERED, and what was your evidence?' [WAIT TIME: 10 seconds.] 'Ochieng and Fatuma — what new question did you add?' [WAIT TIME: 10 seconds.] 'Kamau and Zawadi — which question got your star, and why?' Record any new questions physically on the	DQB Three-Move Protocol (Answer-Evidence / New Question / Star): this metacognitive routine makes learners' growing understanding visible and tracks the storyline arc across all eight lessons. Moving answered questions to a dedicated column gives learners a concrete sense of progress and validates the evidence-gathering sequence. Adding new questions models the scientific disposition that answering one question generates further questions. The re-reading of the driving question keeps the anchoring phenomenon at the centre.	Observable indicator: Each pair has physically moved at least one sticky note to ANSWERED with a written evidence note AND added at least one new question to the board. Teacher reads evidence notes on flipped stickies during circulation — evidence notes should reference a specific salt name and sector (e.g., 'Na_2CO_3 from Lake Magadi is used in laundry — washing soda — which answers why Magadi is commercially valuable despite being caustic'). If a pair cannot move any question to ANSWERED, teacher prompts: 'What did Group D find out about washing soda and soda ash? Does that answer any question about why Lake Magadi is useful?'

	top of the DQB — and asks: 'Which part of the driving question can we now answer more fully than at the start of the unit?'	DQB. Then point to the driving question: 'Let us re-read this together.' [Whole class reads aloud.] 'Can anyone now give a partial answer for the Lake Magadi part of the driving question — specifically about why something caustic can also be commercially useful?' [Cold-call 1–2 learners. WAIT TIME: 15 seconds. Accept partial answers — do not close the question.]		
Phase 5 — Model Building (Community Project Launch + Cumulative Model Revision) 📹 **VIDEO:** [Conservation and the race to save biodiversity] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Hydrolysis of Salts: Equations (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Part A — Cumulative Model Revision (5 minutes): Learners retrieve their personal running model diagram (started in Lesson 1 and revised each lesson). They add a new outer ring or annotation layer labelled 'APPLICATIONS', drawing arrows from each salt they have studied to its real-life sector. They must include at least one arrow that connects back to the Lake Magadi phenomenon and one that connects to the sufuria phenomenon. Part B — Community Project Brief Introduction and Draft (10 minutes): The teacher introduces the sub-strand KICD project: 'Create awareness on the importance of putting required amounts of salt on food to avoid cases of high blood pressure.' Learners receive a Community Sensitisation Message planning frame with four prompts: (1) Who is your audience? (e.g., parents at a school baraza, market traders in Gikomba, farmers in Nakuru County); (2) What is the KEY FACT about salt and blood pressure in one sentence? (3) What local food example will you use to make it real? (e.g., 'When you add extra salt to your ugali or soup…'); (4) What action do you	Part A: 'Open your running model diagram. You have 5 minutes to add the APPLICATIONS layer. Every salt you learned about today must have an arrow pointing to at least one sector. At least one arrow must point back to Lake Magadi and one to the sufuria. Go.' [Circulate. Look specifically for whether learners are connecting Na_2CO_3 to the laundry/Magadi link and $CaCO_3$ to the sufuria scale — these are the two explicit phenomenon threads from the consolidation.] After 5 minutes: 'Hold up your model briefly — I want to see the arrows.' [Quick visual scan of 4–5 models.] Part B: Distribute planning frames. 'This is the start of your community project — the KICD asks you to create awareness about safe salt use and blood pressure. Read the four prompts. You have 8 minutes to draft your first two sentences. Think about who in YOUR community needs this message most — maybe a grandparent who salts their food heavily, or a mama mboga who preserves fish with a lot of salt.' [Allow drafting.] After 8 minutes: 'Share your draft with your partner. Use the checklist. Give one piece of specific praise and one suggestion.' [Allow 2	Cumulative Model Revision (Applications Layer): adding an applications ring to the running model makes the structure-to-use connection explicit and spatial, reinforcing that chemistry knowledge has direct real-world relevance. The model now contains four layers (from Lessons 1–6): acid/base identity → pH measurement → reactions → air-behaviour classification → applications — mirroring the scientific progression from classification to mechanism to use. Community Project Planning Frame: the four-prompt scaffold (audience, key fact, local example, action) teaches science communication as a distinct skill, operationalising the KICD core competency of Communication and Collaboration and the PCI of Self-Efficacy in community health advocacy.	Observable indicator — Model: Teacher's visual scan confirms that learners have added at least 3 labelled arrows with sector names and salt formulae, and that at least one arrow explicitly names Lake Magadi or trona/Na_2CO_3. If arrows are present but formulae are absent, teacher writes on board: 'Arrows need a formula — go back and add it.' Observable indicator — Project Draft: Peer checklist completed (all four items ticked or queried). Teacher collects planning frames as an exit artefact. Success criteria for lesson exit: draft contains (a) a named salt, (b) a named Kenyan food or place, (c) a consequence of excess salt stated in simple terms. Any draft missing two of three criteria is flagged for follow-up at the start of Lesson 7.

	want your audience to take? Learners draft their opening two sentences individually. Pairs peer-review using a simple checklist: Does it name a specific salt? Does it explain the mechanism (osmotic pressure / fluid retention — in simple language)? Does it use a Kenyan food example? Is it respectful and not alarming?	minutes peer review.] Cold-call 2 learners to read their opening sentence: 'Cherotich — read us your first sentence.' [WAIT TIME: 10 seconds.] 'Baraka — yours?' Affirm Kenyan food examples and specific salt names. Close: 'In Lessons 7 and 8, you will refine this message and also investigate rates of reaction — but your community project begins today. Well done.'		

D. TEACHER REFLECTION

After the lesson, reflect on the following: (1) JIGSAW QUALITY — Did every learner in the teaching group genuinely engage with all five sectors, or did some learners disengage while peers were presenting? If disengagement was observed, consider making the Reception Table a graded artefact in Lesson 7 to increase accountability. (2) FORMULA ACCURACY — During the cold-call Master Map construction, how many learners gave a correct chemical formula unprompted? If fewer than half could supply formulae, dedicate 5 minutes at the start of Lesson 7 to a formula flashcard drill. (3) PHENOMENON CONNECTION — When the washing soda–Magadi link was made explicitly, did learners show recognition ('Oh!') or was it new information? If new, the phenomenon has not been sufficiently threaded through Lessons 1–5 and must be re-referenced more deliberately. (4) COMMUNITY PROJECT DRAFTS — Review the collected planning frames before Lesson 7. What is the range of audience choices? Do learners choose local, realistic audiences (baraza, family dinner table) or abstract ones ('all Kenyans')? Concrete local audiences produce more effective sensitisation messages. (5) DQB NEW QUESTIONS — Read all new sticky notes added. Are learners asking mechanism questions ('WHY does salt raise blood pressure?') or only factual ones ('HOW MUCH salt is too much?')? Mechanism questions signal deeper conceptual engagement and should be honoured by a brief teacher response or directed resource in Lesson 7. (6) TIME — The 80-minute double lesson is tightly scheduled. If school operates 40-minute single periods, split at the end of Phase 3 (after the Master Sector Map consolidation), ensuring Phase 4 and 5 open Lesson 7 with a brief recap of the sector findings.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners researched and mapped five real-life sectors (agriculture — CAN/DAP on Rift Valley farms; food industry — NaCl preservation of nyama choma and fish; medicine — ORS, saline, antacid tablets; laundry — washing soda $Na_2CO_3 \cdot 10H_2O$; road use and environment) in which specific named salts play essential roles, and they peer-taught these findings to mixed teaching groups using completed sector maps as visual aids.
What did I learn?	Salts are not just a single substance (table salt / NaCl) but a vast family of ionic compounds produced by neutralisation reactions; each salt's specific properties — solubility, reactivity, hygroscopicity, deliquescence — determine which sector it is used in. Lake Magadi's trona (sodium sesquicarbonate) is the source of industrial soda ash (Na_2CO_3), the same compound as domestic washing soda ($Na_2CO_3 \cdot 10H_2O$), directly linking the anchoring phenomenon to everyday Kenyan laundry practice. Excess dietary NaCl disrupts osmotic balance and raises blood pressure, illustrating that the same salt that is essential for life can be harmful in excess — echoing the phenomenon's central puzzle of 'caustic yet useful.'
How does this explain the phenomenon?	The chemical identity of a salt — its formula, ionic composition, and physical behaviour (including air-behaviour classification from Lesson 5) — determines its real-life application: CAN and DAP release ammonium and nitrate ions that plants need for protein synthesis; antacids contain basic salts ($CaCO_3$, $Mg(OH)_2$) that neutralise excess stomach HCl; washing soda's high carbonate ion concentration softens hard water by precipitating Ca^{2+} and Mg^{2+} ions; and NaCl gritting on cold highland roads lowers the freezing

	point of water by disrupting the ionic lattice of ice — all explained by the same ionic bonding and dissolution principles studied in Lessons 1–5.

LESSON 7 (40 minutes): Driving Question Board Synthesis and Cumulative Model Building	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Students synthesise six lessons of accumulated evidence into a coherent, revised explanatory model that connects acids, bases, salts and their behaviours directly back to the Lake Magadi anchoring phenomenon and the pitted sufuria.
Knowledge	Students can articulate: (i) definitions and examples of acids, bases and salts in terms of H+ and OH- ions and neutralisation products; (ii) the word and balanced equations for neutralisation, carbonate and metal reactions of acids; (iii) the three salt-air behaviours — hygroscopy, deliquescence and efflorescence — with Kenyan examples; (iv) real-life applications of salts across agriculture, food, medicine and industry, including the Lake Magadi trona-to-soda-ash pathway.
Skills	Students demonstrate: constructing and revising scientific models using evidence; evaluating claims on the DQB against experimental data; peer critique using discipline-specific language; written and oral scientific communication.
Attitudes	Students demonstrate intellectual humility by revising earlier predictions, collaborative responsibility in group model building, and appreciation for the relevance of chemistry to Kenyan environments and communities.
Key Inquiry Question	Why are salts important in day-to-day life? How do salts behave when exposed to air? — These two inquiry questions are answered comprehensively by drawing on all prior lesson evidence during the synthesis and model-building process.
Purpose in Storyline	This is the penultimate evidence-gathering lesson. Its role is convergent: students stop collecting new data and instead integrate, evaluate and represent everything learned. It prepares students for the final assessment in Lesson 8 by ensuring every student can explain both anchoring images — Lake Magadi and the pitted sufuria — using full chemical reasoning.
Safety Notes	

B. LESSON OVERVIEW

Lesson 7 opens with a rapid structured revisit of the Driving Question Board, where student groups evaluate every posted question or prediction against the evidence gathered in Lessons 1 to 6. In a 5-minute silent gallery walk followed by group discussion, students categorise DQB entries as Answered, Partially Answered or Still Open. Groups then spend the central block of the lesson constructing a revised cumulative explanatory model — a poster or annotated digital slide — that must include: a pH-scale backbone; acid, base and salt definitions; three reaction types with equations; three salt-air behaviours with Kenyan examples; and an explicit connection back to both anchoring images. Models are shared in a rapid peer-critique gallery walk using a structured feedback protocol. The teacher facilitates a whole-class consensus model drawn live on the board. The lesson closes with students writing one new or lingering question on a sticky note for the DQB, modelling the ongoing nature of scientific inquiry and setting up Lesson 8.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:**	Students enter the classroom to find the full Driving Question Board displayed prominently. Each group	Teacher projects both anchoring images — Lake Magadi and the pitted sufuria — on the screen	Silent individual categorisation activates retrieval of all prior learning before social influence of	Teacher circulates during silent reading and scans notebook entries to identify students who are

Video: pH Scale Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Calculating pH of Salt Solutions (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	(same groups from previous lessons) receives a set of three coloured sticky dots — green (Answered), yellow (Partially Answered), red (Still Open). In silence for 3 minutes, every student individually reads across all DQB entries — predictions from Lesson 1, questions added in Lessons 2 through 6 — and mentally categorises them. Students then record in their notebooks: one entry they are now confident they can fully answer, one they think is only partially answered and one they believe is still genuinely open. This private prediction primes each learner for the subsequent group discussion and prevents groupthink.	before students enter, so they are the first things seen. Teacher says: 'Before we speak today, take 3 minutes to read every card on our Driving Question Board quietly. Ask yourself honestly: can I now explain this using chemistry? Do not discuss yet.' Set a visible 3-minute timer. After the timer, teacher says: 'In your notebook write down one question you can fully answer, one you can partly answer and one you still cannot answer. You have 90 seconds.' WAIT 90 seconds without prompting. Then: 'Hold those thoughts — you will need them very soon.'	group talk occurs. Writing the three categories in the notebook creates a personal evidence audit that will be compared to group consensus, surfacing misconceptions.	unable to name a fully answered question — these students will be paired with stronger peers during the model-building phase. The pattern of red dots placed on the DQB during the subsequent group step will also signal class-wide gaps.
Observe Phase 📹 **VIDEO:** Video: pH Scale Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Calculating pH of Salt Solutions (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Groups of four spend 6 minutes in a structured DQB audit. Each group is assigned a section of the board (acids and pH; reactions of acids; salts and air behaviour; real-life applications). Using their coloured dots, the group places green, yellow or red dots on each entry in their section, then writes a brief evidence justification on a small slip of paper attached beside the entry — citing the specific lesson or experiment that provides the evidence. Groups then do a 2-minute standing gallery walk around the full board to read other groups justifications. Back at their seats, students update their own notebooks: does their personal categorisation from the Predict phase match the group consensus? Where it differs, students note the reason.	Teacher circulates and uses targeted questioning to probe justifications: 'Your group put green on the question about why the sufuria was pitted by lemon juice. Tell me the word equation you would use as evidence.' If a group cannot supply the equation, teacher redirects: 'Look at your Lesson 4 notes — what did dilute HCl do to magnesium ribbon? How is lemon juice similar?' For the gallery walk teacher says: 'You have 2 minutes to read what other groups wrote. Read, do not edit.' WAIT full 2 minutes. After groups return, ask: 'Did anyone find a justification another group wrote that surprised you or that you disagreed with? Hands up.' Take 2 to 3 responses. WAIT 10 seconds between each response before inviting the next.	The evidence-justification slips make reasoning visible and accountable. The gallery walk creates a low-stakes peer-review moment. The notebook comparison step is a metacognitive move — students explicitly notice where personal and social knowledge differ, which is precisely the scientific skill of revising models in light of shared evidence.	Teacher notes which DQB entries collect the most red dots. These become priority content for the whole-class consensus model at the end of the lesson. A cluster of red dots on salt-air behaviour questions, for example, signals that the teacher must revisit hygroscopy and deliquescence terminology during the Explain phase.
Explain Phase 📹 **VIDEO:** Video: pH Scale	Each group now constructs their cumulative explanatory model on A2 paper or a shared digital slide	Teacher circulates continuously during model building, spending approximately 2 minutes per	The five-component mandatory structure prevents superficial listing and requires relational	As teacher circulates, they use a simple 3-column checklist (group name / components present /

Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Calculating pH of Salt Solutions (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	(if devices are available). The model must be organised around five mandatory components displayed as an interconnected diagram, not a list: (1) pH scale backbone labelled with Kenyan examples at key points — lemon juice at pH 2, Lake Magadi water at pH 10 to 11, uji at pH 6; (2) definitions of acid, base and salt using H+ and OH- ion language; (3) three reaction types with at least one word equation each — neutralisation, carbonate reaction, metal reaction — each annotated with the Kenyan context in which it occurs; (4) the three salt-air behaviours with one Kenyan example each — NaCl in Mombasa coastal humidity, CaCl2 in industrial desiccants, Na2CO3 10H2O efflorescence on the sufuria scale; (5) a central image or symbol of both anchoring phenomena — Lake Magadi trona crusts and the pitted sufuria — with arrows showing how the chemistry concepts connect to each image. Groups have 12 minutes to build their model. A sentence-starter scaffold is provided on the board: 'We know this because in Lesson ___ we observed/measured/read that ___.' This scaffold forces every claim to be evidence-anchored.	group. At each group teacher first listens for 20 seconds before speaking — this models scientific patience and gives students time to articulate their thinking. Prompts for groups that are listing rather than connecting: 'Your model shows these as five separate boxes. Where does the arrow go from neutralisation to salt formation to what happens in the sufuria? Draw that connection.' For groups that have completed a strong draft early: 'You have connected lemon juice to the sufuria pitting — can you now show on your model why the same lemon juice is safe to drink? What stops it from damaging your stomach lining the way it damaged the aluminium?' WAIT for student response before scaffolding further. For the trona connection, if a group has not included it, say: 'Lake Magadi has not appeared on your model yet. Trona is sodium sesquicarbonate — a salt. Which of your five components should it appear in? Where does it fit?' WAIT 15 seconds.	thinking. The sentence-starter scaffold embeds evidence norms. The teacher questioning strategy of listening before speaking models scientific discourse. Requiring both anchoring images to appear on the model ensures the lesson returns to the phenomenon and does not drift into abstract chemistry.	connections evident) to note which groups have all five components and which are missing connections. This data directly informs how much time teacher spends at each group and which gaps to address in the consensus model phase.
Driving Question Board (DQB) Creation 📹 **VIDEO:** Video: pH Scale Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos	Groups post their completed models on the classroom walls for a rapid 4-minute peer-critique gallery walk. Each student carries two sticky notes — one green, one orange. On the green note they write one specific strength of a model that is not their own: something that is scientifically correct, well connected or clearly explained with evidence. On the	Before the gallery walk, teacher says: 'Your feedback must be specific and evidence-based. Do not write good job. Write exactly what is correct and why, or exactly what is missing and why it matters. You have 4 minutes to visit at least two models that are not yours.' Set a visible 4-minute timer. WAIT the full time without intervening in student discussions at the models	The two-colour sticky note protocol ensures feedback is balanced — strengths before challenges — which supports a safe intellectual climate while maintaining rigour. Requiring students to select one orange-note challenge for their presentation gives agency and prepares for the Explain phase presentations. The new DQB entries maintain the	Teacher photographs the new DQB section (What we still want to know) immediately after it is populated. These questions are analysed to confirm whether the Lesson 8 environmental impact focus on fertiliser salts and eutrophication is already present in student thinking, or whether teacher will need to introduce that context more explicitly.

📖 **HTML:** [Calculating pH of Salt Solutions (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	orange note they write one specific question or friendly challenge: something they think is missing, unclear or that they would like the group to justify further. Students post both notes directly on the relevant model. After the gallery walk, each group has 90 seconds to read the feedback on their own model and identify the single most important orange-note challenge they will incorporate into their verbal presentation. Students also write one new or remaining question of their own on a fresh sticky note and add it to the DQB under a new section labelled 'What we still want to know before Lesson 8.' This section will be the entry point for the environmental impact investigation in Lesson 8.	— allow authentic peer dialogue. After groups read their feedback, teacher says: 'You have 90 seconds to identify the one orange-note challenge your group will respond to in your presentation. Just one — choose the most scientifically important one.' WAIT 90 seconds. Then for the new DQB entries: 'Before we hear presentations, write one question you still genuinely have — something this unit has not yet fully answered for you — and add it to the board. It can be about Lake Magadi, about salt and health, about the environment, about anything our evidence has not closed. Your question is valuable data for science.'	epistemological norm that science generates new questions, and they create a direct bridge to Lesson 8 eutrophication content.	
Model Building Phase 📹 **VIDEO:** [Video: pH Scale] Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** [Calculating pH of Salt Solutions (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Each group has 90 seconds to present their model to the class, explicitly responding to the orange-note challenge they identified. Presentation structure is fixed: (1) Here is how our model explains the Lake Magadi image; (2) Here is how our model explains the pitted sufuria; (3) Here is the challenge we received and how we respond to it. After all presentations, the teacher draws a consensus class model live on the board, calling on specific students to contribute each component: 'Amara, which pH value did we measure for Lake Magadi water in Lesson 2 and what does that tell us about its H+ concentration?' 'Kiprotich, give me the word equation for what lemon juice did to the aluminium sufuria.' 'Wanjiru, which salt-air behaviour explains the white crusts on the sufuria walls from hard water?' The consensus model is photographed	During group presentations teacher enforces the 90-second time limit using a visible timer — this develops disciplined scientific communication. After each presentation, teacher pauses for exactly 10 seconds before commenting — this WAIT TIME signals that peer responses are welcome before teacher evaluation. When building the consensus model on the board, teacher says: 'I am not drawing my model. I am drawing what your evidence tells us. If I write something you disagree with, say so and give me the evidence.' This positions the teacher as a facilitator of consensus, not an authority delivering truth. For the closing reflection, teacher says: 'You have 3 minutes. Open your Lesson 1 notebook entry — your very first prediction about Lake Magadi and the sufuria. Read it. Then write two sentences: what	The fixed three-part presentation structure ensures every group connects back to the phenomenon and engages with critique. Live co-construction of the consensus model on the board is a high-leverage sensemaking move — it makes class knowledge a shared public artefact rather than six separate group artefacts. The closing two-sentence reflection is a metacognitive consolidation task that makes learning visible to both student and teacher. The teacher closing statement creates explicit narrative continuity into Lesson 8, maintaining storyline coherence.	The two-sentence closing reflection is collected as an exit task — teacher reads these before Lesson 8 to identify students who are still holding Lesson 1 misconceptions about acids or salts. The consensus model on the board serves as a public benchmark: any student whose group model diverges significantly from the consensus model is identified for targeted support before the written assessment in Lesson 8.

	and shared with students as the class reference for Lesson 8 revision. Lesson closes with teacher returning to the anchoring images and asking: 'Six lessons ago you predicted what was happening in these two images. Look at your Lesson 1 prediction in your notebook. How has your thinking changed? Write two sentences comparing your original prediction to your model today.'	you got right, and what you now understand that you did not understand then. Be honest — changing your mind because of evidence is exactly what scientists do.' WAIT the full 3 minutes in silence. Do not ask students to share unless a volunteer offers — this protects intellectual safety around admitting prior misconceptions. Close by saying: 'In Lesson 8 we will take one of the questions from the new section of our DQB — about what happens when too many fertiliser salts enter a lake — and investigate whether chemistry that is useful on a Rift Valley farm can become harmful in a Kenyan lake. The same question we started with: caustic yet useful, essential yet dangerous.'		

D. TEACHER REFLECTION

After the lesson, consider the following questions: Did every group successfully place both anchoring images at the centre of their model, or did some groups treat them as decorative rather than explanatory? If groups struggled to connect trona to the soda ash industrial pathway established in Lesson 6, what additional bridging question could be asked at the start of Lesson 8? Which new DQB questions posted in the What we still want to know section were closest to the Lesson 8 eutrophication content, and how can those student questions be used as the opening hook rather than a teacher-introduced phenomenon? Were there students who remained silent during all group and class discussion phases? What structural change — paired talk, written contribution before oral, role assignment — could ensure those voices are present in Lesson 8? Finally, does the consensus class model accurately represent all five required components? If any component is weak, plan a 3-minute targeted revisit at the very start of Lesson 8 before introducing new content.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Students observed the full set of Driving Question Board entries accumulated since Lesson 1, peer models from all groups during the gallery walk, and the live consensus model built collaboratively on the board at the end of the lesson. They also observed their own Lesson 1 written predictions compared to their current understanding.
What did I learn?	Students learned that all six lessons of evidence converge on a coherent explanatory framework: the pH scale, acid-base-salt definitions, three reaction types, three salt-air behaviours and real-life applications all connect to the two anchoring images. They learned that scientific models are revised iteratively as evidence accumulates, and that Lake Magadi trona is chemically the same family of compound as the mineral scale in the sufuria and the washing soda in domestic laundry — illustrating that one phenomenon can be explained by multiple intersecting chemistry concepts.
How does this explain the	Students explained: why Lake Magadi water is caustic (high pH, high OH- concentration from carbonate and bicarbonate salts) yet

phenomenon?	yields commercially and domestically useful soda ash; why lemon juice pitted the aluminium sufuria overnight (acid-metal reaction producing aluminium salt and hydrogen gas) but is safe to consume in dilute form; why the sufuria scale formed (dissolved mineral salts in hard water deposited on evaporation — salt-air and solubility behaviour); and how every phenomenon in the unit is an instance of the same underlying chemistry of acids, bases and salts acting on materials and living systems in the Kenyan environment.

LESSON 8 (80 minutes): Final Explanation: Environmental Impact, Community Project Showcase & Assessment — Closing the Loop on Lake Magadi and the Sufuria	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To synthesise all sub-strand learning by linking eutrophication at Lake Victoria to excess fertiliser salts, showcasing community sensitisation projects, and completing a written formative assessment that uses the anchoring phenomenon as stimulus — demonstrating full command of acids, bases and salts chemistry.
Knowledge	Students will know: (h) how to classify salts according to their behaviour when exposed to air (hygroscopy, deliquescence, efflorescence); (e) how to outline applications of salts in real-life situations (agriculture, food industry, medicine, laundry, road use); (f) how excess fertiliser salts cause eutrophication and environmental harm; (b) the role of acids and bases in biological and chemical processes; (c) the products of chemical reactions of acids with metals, carbonates and bases.
Skills	Students will be able to: carry out and present findings on chemical properties of dilute acids and bases; investigate and explain the behaviour of salts exposed to air; discuss the effect of salts on the environment (eutrophication, soil and air pollution); sensitise the community on the acidity and basicity of household substances and correct salt use; apply sub-strand knowledge to explain the anchoring phenomenon (Lake Magadi crusts, pitted sufuria, clumping canteen salt) in writing.
Attitudes	Students will appreciate: the ongoing nature of scientific inquiry; the responsibility of every citizen to use salts and acids safely and sustainably; the importance of community education in preventing environmental harm from fertiliser overuse; and the value of evidence-based explanations over guesswork.
Key Inquiry Question	1. Why are salts important in day-to-day life? 2. How do salts behave when exposed to air?
Purpose in Storyline	This final lesson closes the storyline arc that opened in Lesson 1 with the two puzzling images. Students now possess all the chemical tools needed to fully explain: (i) why Lake Magadi's trona crusts form and why they are both caustic and commercially valuable; (ii) why lemon juice pitted the aluminium sufuria overnight; (iii) why canteen salt clumps in the rainy season (hygroscopy/deliquescence); and (iv) how fertiliser salts washing into Lake Victoria trigger the water hyacinth eutrophication crisis. The lesson also surfaces any remaining questions, modelling the idea that science never fully closes — it only deepens.
Safety Notes	No new hazardous chemicals are used in this lesson. If any reagents are on display for the community project showcase, ensure stoppered bottles are used, labels are clearly visible, and no students handle concentrated acids or bases without gloves. Written assessment papers should not contain actual chemical samples. Remind students that Lake Magadi water (pH ~10–11, high Na_2CO_3) is caustic and should never be tasted or handled without protection — relevant when discussing the anchoring phenomenon.

B. LESSON OVERVIEW
Lesson 8 is the capstone of Sub-Strand 2.4. It unfolds in five tightly linked phases: (1) a structured Predict phase where students revisit their original L1 predictions about the anchoring phenomenon and rate their own understanding growth; (2) an Observe phase built around a short eutrophication case study on Lake Victoria's water hyacinth crisis, connecting excess fertiliser salts to real environmental harm; (3) an Explain / Final Explanation phase in which student groups present their community sensitisation projects and the class collectively constructs one final, comprehensive explanation of every element of the anchoring phenomenon; (4) a Driving Question Board (DQB) completion ceremony where all previously posted questions are answered or flagged as 'still open', and each student adds one remaining question on a sticky note; and (5) a Model Building phase in which students compare their original L1 conceptual model with their final revised model, annotating the differences in colour. The lesson closes with a 25-minute written formative assessment using the anchoring phenomenon images as stimulus. Throughout, the teacher uses Socratic

questioning, cold-calling, think-pair-share and whole-class gallery walks. All Kenya-relevant contexts (Lake Magadi, sufuria, ugali pot, canteen salt, Rift Valley soda ash industry, Lake Victoria water hyacinth) are foregrounded.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Phase 1 — Predict: Revisiting Original Predictions & Self-Rating Understanding Growth 📹 **VIDEO:** [Proof: Rhombus diagonals are perpendicular bisectors] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Properties of Acids (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Students receive back their original L1 prediction sheets (kept by the teacher since Lesson 1) showing what they wrote about the two anchoring phenomenon images — Lake Magadi's soda crusts and the pitted aluminium sufuria. Working individually for 4 minutes, they re-read their L1 predictions and then on a sticky note write: (a) one thing they predicted correctly, (b) one thing they got completely wrong, and (c) one chemical term they now use confidently that they did not know in Lesson 1. Students then do a brief 'fist-to-five' self-rating (0 fingers = I still cannot explain the phenomenon; 5 fingers = I can fully explain both images using acid, base and salt chemistry). The teacher records the class distribution on the board. Pairs share their 'one wrong prediction' with a neighbour for 90 seconds, explaining what they know now that corrects it. The class hears three cold-called shares.	Before the lesson, distribute returned L1 prediction sheets face-down on desks. Say: 'Do not flip them until I say go.' After 30 seconds of suspense: 'Flip them now. Read what you wrote eight lessons ago. You have 4 minutes — no talking.' [WAIT 4 minutes]. Then: 'On a sticky note, write three things: one prediction you got right, one you got badly wrong, and one chemical term you now own that you did not know in Lesson 1.' [WAIT 3 minutes]. 'Hold up your fists — show me zero to five fingers: how well can you NOW explain the Lake Magadi image and the pitted sufuria?' Scan the room and narrate: 'I see mostly fours and fives — we have come a long way.' Cold-call 3 students: 'Amina — what was your most wrong prediction and what do you know now?' [WAIT 10 seconds before prompting]. 'Kipchoge — same question.' 'Wanjiru — what chemical term do you now own?' Record key terms students name (deliquescence, efflorescence, hygroscopy, neutralisation, pH, eutrophication) on the board as a 'vocabulary harvest'. Say: 'Keep your prediction sheets — you will need them in the assessment.'	Metacognitive Reflection via Prediction Audit — by comparing their original naive predictions with current understanding, students activate prior knowledge, surface remaining gaps, and build epistemic confidence. The fist-to-five provides the teacher with rapid formative data to calibrate pacing for the rest of the lesson. The vocabulary harvest makes implicit learning explicit and public.	Observable indicator: All students can name at least one chemical term they now use confidently that was absent from their L1 prediction. Teacher notes any student showing 0–2 fingers on the fist-to-five and proactively pairs them with a stronger peer during Phase 3. The three cold-called responses reveal whether students can correctly identify their own misconceptions — a higher-order metacognitive skill.
Phase 2 — Observe: Eutrophication Case Study —	Students read a one-page illustrated case study (provided on the ARES platform) titled 'From Fertiliser Bag to Floating Weed:	Distribute or display the case study. Say: 'You have 8 minutes to read carefully and answer the three questions in your exercise	Evidence-to-Explanation Bridge using Real Data — students move from observable data (seasonal nitrate concentrations in a data	Observable indicator: Students correctly identify CAN and DAP as salts (ionic compounds) in Q1. In Q2, students use the data table to

Lake Victoria's Water Hyacinth Crisis 📹 **VIDEO:** Proof: Rhombus diagonals are perpendicular bisectors Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Properties of Acids (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	How Salts Are Choking Lake Victoria.' The text and labelled diagram describe: (i) how farmers in the Lake Victoria basin (Kisumu, Homa Bay, Siaya counties) apply nitrogen-rich fertilisers (calcium ammonium nitrate — CAN, diammonium phosphate — DAP) to their maize and sugarcane fields; (ii) how heavy rains wash nitrate and phosphate salts into streams and finally into the lake — a process called nutrient run-off; (iii) how elevated dissolved nutrients cause explosive algal blooms and water hyacinth (Eichhornia crassipes) growth, which depletes dissolved oxygen, kills fish, and blocks boat routes for Luo fishing communities; (iv) a data table showing nitrate concentration (mg/L) at three lake monitoring stations (Kisumu Port, Dunga Beach, Kendu Bay) across two seasons. Students answer three guided questions in their exercise books: Q1 — Which chemical family do the fertiliser compounds (CAN, DAP) belong to, and how do they behave when dissolved in water? Q2 — Using the data table, describe the trend in nitrate concentration between the dry season and the rainy season, and suggest a reason. Q3 — How does this eutrophication phenomenon connect to the Lake Magadi anchoring phenomenon — what do trona, CAN and DAP all have in common chemically?	book. This is silent individual work.' [WAIT 8 minutes. Circulate, note common errors, especially students who confuse eutrophication with acidification]. After 8 minutes: 'Pause writing. Turn to your neighbour and compare answers to Q3 — that is the hardest one. You have 2 minutes.' [WAIT 2 minutes]. Cold-call 3 pairs: 'Otieno and Aisha — what did you agree on for Q3?' [WAIT 10 seconds]. After each response, press: 'Can you name the chemical family AND one property that makes these salts soluble in water and available for run-off?' Scribe key phrases on the board: 'ionic compounds', 'soluble in water', 'dissociate into ions', 'nitrogen and phosphate ions are plant nutrients'. Then bridge to the phenomenon: 'So the same chemistry that makes salts useful in agriculture — solubility, ion release — also makes them dangerous when overused. Just like the trona at Magadi is both commercially valuable AND caustic to flamingo skin. Chemistry is context.'	table) to causal explanation (salt solubility → ion run-off → eutrophication). Linking this to the anchoring phenomenon (trona = salt, CAN = salt, both useful and harmful depending on context) deepens the cross-cutting concept of cause-and-effect and connects environmental science to the chemistry content. The pair-share on Q3 surfaces reasoning before whole-class discussion.	describe the seasonal trend accurately (higher nitrate concentration in rainy season due to run-off). In Q3, students articulate that trona, CAN and DAP are all salts — ionic compounds that dissolve in water and release ions. Teacher circulates during individual work and targets students who have not connected eutrophication to salt chemistry with a guiding question: 'What category of compound is calcium ammonium nitrate? What does it do when it dissolves in water?'
Phase 3 — Explain: Final Explanation — Community Project Showcase & Comprehensiv	This phase has two parts. PART A (Community Project Showcase, 18 minutes): Student groups (formed in Lesson 6) take turns presenting their community sensitisation projects on the acidity and basicity of household substances and	PART A: Announce group order (pre-assigned). 'Each group: 3 minutes, no more. Audience: complete your peer-feedback card as you listen. I am listening for correct use of these terms —' [point to vocabulary harvest on	Claim–Evidence–Reasoning (CER) Framework for Final Explanation — CER is a structured argumentation strategy that forces students to separate observations (evidence) from interpretations (claims) and justify with chemical	Observable indicators: (1) During showcase, each group uses at least three correct chemical terms from the sub-strand vocabulary list. Teacher notes groups that use terms incorrectly and corrects immediately with a class-wide

e Phenomenon Explanation 📹 **VIDEO:** [Proof: Rhombus diagonals are perpendicular bisectors] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Properties of Acids (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	correct salt use. Each group has 3 minutes maximum. Presentations may take the form of a poster, a short drama/role-play, a pamphlet, or a digital slide (on ARES). Audience members use a peer-feedback card with three sentence starters: 'One chemistry fact I heard correctly used was…', 'One thing I would add to make this more accurate is…', 'One community safety message I will personally share is…'. PART B (Final Phenomenon Explanation, 12 minutes): The whole class works together to construct one comprehensive, written explanation of all three anchoring phenomenon observations — (i) Lake Magadi's white trona crusts and caustic water, (ii) the pitting of the aluminium sufuria by lemon juice, (iii) clumping of canteen salt in the rainy season — using a structured 'Claim–Evidence–Reasoning' (CER) framework on a large class chart paper. The teacher scribes. Students contribute statements and are challenged to cite the lesson number (evidence lesson) from which each claim is drawn.	board]. Use a visible 3-minute timer. After each presentation, give one specific affirmation: e.g., 'Group 2 — you correctly described sodium chloride as a neutral salt that is hygroscopic under very humid conditions. Well done.' Gently correct errors on the spot: 'Group 4 — you said lemon juice is a base. Which lesson gave us the pH of lemon juice? Waweru, can you help?' [WAIT 10 seconds]. PART B: Display the two anchoring phenomenon images on the screen again — the same images from Lesson 1. Say: 'We are now going to write the explanation we could not write in Lesson 1. I will scribe. You will speak. Use Claim–Evidence–Reasoning.' Guide with prompts: 'What CLAIM can we make about the trona at Magadi?' [WAIT 10 seconds, cold-call]. 'What EVIDENCE from our lessons supports that claim?' [WAIT 10 seconds]. 'What REASONING connects the evidence to the claim?' Repeat for sufuria pitting and canteen salt clumping. Ensure the final CER chart explicitly names: Na_2CO_3 (trona/sodium sesquicarbonate), alkaline pH, efflorescence; Al + dilute acid (lemon juice/citric acid) → aluminium salt + H_2; hygroscopy/deliquescence of NaCl under high humidity.	reasoning. Applying it to the anchoring phenomenon creates a public, co-constructed record of the class's collective understanding across all eight lessons. The community project showcase activates the KICD Communication and Collaboration competency and the Self-efficacy competency, while also fulfilling the KICD mandated project: 'Sensitise community on the acidity and basicity of substances used in day-to-day life.'	question. (2) During CER chart construction, students spontaneously cite lesson evidence (e.g., 'In Lesson 3 we tested lemon juice with universal indicator and got pH 2–3'); this citation behaviour indicates deep retention. (3) The completed CER chart is photographed and uploaded to ARES as a shared class resource. Any gap in the CER chain (missing evidence or reasoning) is flagged by the teacher as a question for the DQB.
Phase 4 — Driving Question Board (DQB) Completion: Closing Questions & Honouring What Remains	The class DQB (which has been building since Lesson 1) is displayed prominently. The teacher reads each posted question aloud. For each question, students vote with thumbs: thumb up = 'we can now answer this'; thumb sideways = 'we can partially answer this'; thumb down = 'this is	Stand at the DQB. Say: 'This board has been our scientific journal for eight lessons. Let us see how far we have come.' Read each question slowly and dramatically. After each one: 'Thumbs — up, sideways or down?' Scan the room. For 'thumb up' questions, cold-call: 'Fatuma —	DQB Closure and 'Still Wondering' Protocol — this strategy makes visible the epistemological journey from naive curiosity (L1) to informed curiosity (L8). Distinguishing between 'answered', 'partially answered', and 'still open' questions teaches students that science is a process	Observable indicator: Each student posts a genuine, chemically-grounded 'Still Wondering' question — not a repeat of content already covered. The teacher scans posted questions for scientific quality. High-quality questions (e.g., 'Why does trona form at Lake Magadi

Unknown 📹 **VIDEO:** [Proof: Rhombus diagonals are perpendicular bisectors] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Properties of Acids (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	still genuinely open.' The class collectively answers the 'thumb up' questions aloud, one by one, with cold-called students providing the answers. 'Thumb sideways' questions are annotated on the DQB with a brief partial answer. 'Thumb down' questions are left intact and circled in a different colour — honoured as evidence of the frontier of knowledge. Each student then individually writes one remaining question they still have on a fresh sticky note — it must be a genuine question that arose from the lessons, not a repetition of an already-answered DQB question — and posts it on a designated 'Still Wondering' section of the DQB. Students share their sticky note questions with a partner before posting.	answer this one fully, in one sentence, using chemical vocabulary.' [WAIT 10 seconds]. Affirm correct answers and annotate the DQB card with a green tick. For partially answered questions, say: 'Good. We know [X] but we do not yet know [Y]. That is honest science.' For genuinely open questions, say: 'This question is still open. Scientists at the Kenya Marine and Fisheries Research Institute in Kisumu are still investigating questions like this about Lake Victoria. Your curiosity is legitimate.' Distribute sticky notes. Say: 'Now write YOUR remaining question. It must be genuine — something that still puzzles you after eight lessons. You have 2 minutes.' [WAIT 2 minutes]. 'Share it with your partner. Does your partner think it is a real, answerable scientific question, or is it philosophical? Adjust if needed. Now post it on the Still Wondering section.' After posting: 'Look at our Still Wondering wall. This is what science looks like — we never fully close the question. Every answer opens three more.'	of progressive refinement, not a finished product. The 'Still Wondering' sticky notes honour intellectual humility and model the practice of real scientists. This directly addresses the KICD competency of Learning to Learn and the value of Responsibility.	but not at Lake Naivasha, even though both are soda lakes?' or 'Could the efflorescence from washing soda be recollected and reused?') are highlighted to the class as examples of excellent scientific thinking. Questions that reveal persistent misconceptions (e.g., 'Is acid always dangerous?') are gently reframed as teaching moments.
Phase 5 — Model Building: L1 vs Final Model Comparison & Written Formative Assessment 📹 **VIDEO:** [Proof: Rhombus diagonals are perpendicular bisectors] Source: Khan Academy (English - US curriculum)	PART A — Model Comparison (10 minutes): Students receive back their original Lesson 1 conceptual model drawings (kept by the teacher) showing their initial understanding of what was happening at Lake Magadi and in the pitted sufuria. Using a green pen or pencil, they annotate their original model with corrections and additions representing their final understanding. They must add: (1) the name and formula of the key compound at Lake Magadi (trona / sodium sesquicarbonate,	Distribute original L1 model drawings. Say: 'You have 10 minutes. Use a green pen or pencil only — so your original thinking stays visible. Annotate, correct, add. This is not about being right in Lesson 1 — it is about showing how far you have travelled.' [WAIT 10 minutes. Circulate silently. Do not intervene unless a student is genuinely stuck — offer only: 'Look at your CER chart on the board.']. After 10 minutes: 'Stop. Keep your model face up on your desk — you will refer to it during the	Annotated Model Comparison (L1 vs Final) — by physically marking up their original model in a contrasting colour, students make their conceptual growth tangible and visible. This is a powerful metacognitive strategy that consolidates learning across all eight lessons. The written formative assessment using the anchoring phenomenon as stimulus is the highest-fidelity assessment possible: it requires students to transfer chemistry knowledge to the exact context	Observable indicators: (1) In Part A, annotated models show at least three substantive changes from the L1 original, using correct chemical terminology and at least one balanced equation. (2) In Part B written assessment, the teacher uses the following marking criteria — Q1: names sodium carbonate / trona, identifies alkaline nature (high pH) and efflorescence/crystallisation on evaporation (2+2); Q2: correct balanced equation $2Al + 6HCl \rightarrow 2AlCl_3 + 3H_2$ AND correct

🔍 Search ARES for similar videos 📖 **HTML:** [Properties of Acids (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	$Na_3H(CO_3)_2 \cdot 2H_2O$, or simplified as Na_2CO_3); (2) the chemical reaction equation showing aluminium reacting with citric acid (lemon juice); (3) the correct salt behaviour label for canteen salt in humid conditions (hygroscopic); (4) an arrow linking fertiliser run-off to eutrophication. Students write two sentences below their model: 'My model changed most in understanding [X] because…' PART B — Written Formative Assessment (25 minutes, individual, silent): Students answer four structured questions using the anchoring phenomenon images as stimulus. Q1 (4 marks): Identify the type of salt found at Lake Magadi and explain TWO properties that explain why it accumulates as white crusts on the shoreline. Q2 (4 marks): Write a balanced chemical equation for the reaction between aluminium and dilute hydrochloric acid. Using this as a model, explain why lemon juice (citric acid) pitted the inside of the aluminium sufuria overnight. Q3 (4 marks): A canteen cook in Nairobi notices that the salt in the shaker clumps together every April during the long rains. Name the property of salt responsible for this and explain, using the concept of ions and water molecules, how clumping occurs. Q4 (4 marks): Explain how excess use of CAN fertiliser on maize farms near Kisumu can lead to the death of Nile tilapia (ngege) in Lake Victoria. Use the terms: salt, solubility, run-off, eutrophication, dissolved oxygen in your answer.	assessment if needed.' Distribute assessment papers. Say: 'This is individual, silent work. You have 25 minutes. The images on your paper are the same two images from Lesson 1. Use everything you know. Write in full sentences. Chemical equations must be balanced.' [WAIT 25 minutes. Circulate, do not assist.] At 25 minutes: 'Pens down. Pass your paper to the front.' Close the lesson: 'In Lesson 1, you looked at these two images and could only ask questions. Today you explained them. That is what science education does — it transforms confusion into understanding. Well done.' [Allow 10 seconds of genuine silence before dismissal — a moment of collective recognition].	that opened the unit, demonstrating genuine understanding rather than rote recall. The four questions map directly to the KICD specific learning outcomes (h, e, f, b, c) and the two key inquiry questions.	explanation linking citric acid's H^+ ions to oxidation of aluminium surface (2+2); Q3: names hygroscopy or deliquescence, explains that Na^+ and Cl^- ions attract polar water molecules from humid air, causing clumping (2+2); Q4: mentions nitrate/phosphate ions from CAN dissolving and running off into lake water, causing algal bloom, decomposition depletes dissolved oxygen, ngege suffocate (4 marks for complete causal chain). Scripts are marked before Lesson 9 of the next sub-strand and returned with written feedback.

D. TEACHER REFLECTION

After the lesson, reflect on the following questions and record your notes in the ARES teacher journal: 1. PHENOMENON CLOSURE: Were students able to fully explain both anchoring phenomenon images using chemical vocabulary by the end of Phase 3? If not, which specific element (Lake Magadi chemistry, sufuria pitting, salt clumping, or eutrophication) remained unclear — and what would you teach differently in a future iteration? 2. COMMUNITY PROJECT QUALITY: Did the community sensitisation presentations demonstrate genuine understanding of acid-base chemistry, or were they superficial? If groups struggled, consider building more structured project scaffolding into Lessons 6 and 7 in future cycles. 3. ASSESSMENT DATA: Review the written formative assessment scripts before the next lesson. Tally: how many students correctly answered all four questions? How many scored on Q2 (balanced equation)? If fewer than 60% balanced the equation correctly, plan a brief equation-balancing consolidation activity at the start of Sub-Strand 2.5 Rates of Reactions. 4. DQB STILL WONDERING: What were the highest-quality 'Still Wondering' questions posted by students? Could any of these become the entry phenomenon for Sub-Strand 2.5 (Rates of Reactions)? For example, a question like 'Does the temperature of Lake Magadi's water affect how fast trona crystals form?' bridges beautifully into reaction rates. 5. EQUITY CHECK: Did all students — including those who showed 0–2 fingers on the fist-to-five in Phase 1 — participate meaningfully in at least one phase of the lesson? If not, what differentiation strategy would better support them in a capstone lesson? 6. KICD COMPETENCY REALISATION: The community sensitisation project is the primary vehicle for the KICD mandated Communication and Collaboration and Self-efficacy competencies in this sub-strand. Did students demonstrate confidence in public communication of scientific ideas? Note specific examples for learner profile records.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Students observed: (1) a case study with real seasonal nitrate concentration data from three Lake Victoria monitoring stations (Kisumu Port, Dunga Beach, Kendu Bay), showing higher values in the rainy season due to fertiliser run-off; (2) their own original Lesson 1 prediction sheets and conceptual model drawings, revealing how their understanding has grown; (3) the two anchoring phenomenon images (Lake Magadi soda crusts; pitted aluminium sufuria) displayed again for final explanation; (4) peer community sensitisation presentations (posters, dramas, pamphlets) on household acid-base chemistry and correct salt use.
What did I learn?	Students learned: (1) excess fertiliser salts (CAN, DAP) dissolve in water, run off into Lake Victoria, cause eutrophication (algal bloom → oxygen depletion → fish death) — directly linking salt solubility and ion chemistry to environmental harm; (2) trona (sodium sesquicarbonate / Na_2CO_3) at Lake Magadi is an alkaline salt that crystallises by efflorescence as lake water evaporates in the Rift Valley heat; (3) lemon juice (citric acid, low pH) reacts with aluminium metal (Al + acid → aluminium salt + H_2), pitting the sufuria; (4) canteen salt clumps in April rains because NaCl is hygroscopic — Na^+ and Cl^- ions attract polar water molecules from humid air; (5) their own conceptual models changed substantially across eight lessons, and scientific understanding grows by revising prior models with new evidence.
How does this explain the phenomenon?	Students explained, using the Claim–Evidence–Reasoning framework: (1) WHY Lake Magadi's trona forms white shoreline crusts — evaporation concentrates Na_2CO_3 dissolved in the alkaline lake water, which then crystallises (efflorescence) as the water dries; the high pH makes the water caustic yet commercially valuable as soda ash for the Magadi Soda Company; (2) WHY lemon juice pitted the aluminium sufuria — citric acid (H^+ ions, pH 2–3) reacted with the aluminium metal surface overnight via an acid-metal

	reaction, producing aluminium salt and hydrogen gas, corroding the pot wall; (3) WHY canteen salt clumps in the long rains — sodium chloride is hygroscopic, absorbing water vapour from humid April air onto its ionic crystal surfaces; (4) HOW fertiliser salt overuse harms Lake Victoria — nitrate and phosphate ions from CAN and DAP run off into the lake, fuel explosive algal and water hyacinth growth, decomposing biomass depletes dissolved oxygen, and Nile tilapia (ngege) suffocate — all traceable to the solubility and ionic nature of salts studied throughout the sub-strand.

DIFFERENTIATION AND INCLUSION		
Learner Need	**Support Strategies**	**Extension Strategies**
English Language Learners	Provide bilingual glossaries; allow use of first language for initial model drawing.	Write extended explanations of observations; lead group discussions in English.
Students with learning difficulties	Provide structured note frames; pair with a supportive partner during investigations.	Mentor peers; create additional visual models to explain concepts.
Gifted and advanced learners	Access to additional reading materials; challenge questions during DQB.	Lead model-building discussions; research and present extension topics to class.